Supplementary Information: Genome-scale context, stoichiometric refinement, and neuro-symbolic reasoning for proteome function annotation

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S1 Benchmark suite overview, Stage 0 calibration, and reference panels

Supplementary Table S1 lists every benchmark reported in the main text and in this supplement, with the cohort that was run, the Stage 0 evidence channels, the pipeline stages and command-line options used, the comparators, the result files in `results/`, and the reproduction pipeline under `paper-reproducibility/pipelines/`. Stage 1 priors and Stage 2A enforcement run inside `gspa integrate`; Stage 2A is switched on with `-enforce-stage2` (with `-enforce-consistency-mode remove` or `relax`), and the reaction-local genomic context suggester with `-rlc-suggester`. The integrator also exposes `-enable-fba-refiner` (Stage 2B) and `-enable-agent` (Stage 3), but no reported benchmark uses them: Stage 2B runs through the Python refiner `scripts/stoichiometric_locus_refiner.py` and Stage 3 through the agent harness on the integrator's anomaly records. In the whole-genome CAFA benchmark, Stage 0 to Stage 2B are implemented inside `scripts/run_cafa_whole_genome_splits.py` over the

benchmark's own retrained channels, and in the Price-149 and BacDive benchmarks the GSPA arms are computed by their evaluation scripts.

Table S1. Benchmarks, cohorts, Stage 0 channels, stages run, comparators, result files, and reproduction pipelines. Result files are under `results/`. Pipeline identifiers refer to `paper-reproducibility/pipelines/`; P01, P06, and P09 are listed in `paper-reproducibility/README.md` but have no folder yet, and their scripts are named in the table. "Pending" marks benchmarks whose regeneration has not finished. Table and figure numbers without the prefix S refer to the main text.

Benchmark (manuscript item)	Cohort	Stage 0 channels	Stages run (options)	Comparators	Result files	Pipeline
Whole-genome CAFA splits (Table 1; Table S4)	6 complete prokaryotic genomes (20,697 coding sequences); 3,315 experimental targets, 1,060 of them post-t_0; 2 time and 4 homology splits	6 channels retrained per split: DIAMOND k-NN, InterPro-LR, DeepGOCNN, ESM-2 3B k-NN, ESM-2 3B MLP, ESM-2 3B C-HMCNN	Stage 0, Stage 1 (B_1–B_6), 2A, 2B, implemented in the benchmark script	14 retrained per-protein baselines (6 re-implementations on ESM-2 3B embeddings); pretrained DeepFRI and DeepGO-SE; GSPA-Seq	`cafa_whole_genome_time_and_homology_splits.tsv`, `cafa_whole_genome_per_species.tsv`	P01 (`scripts/run_cafa_whole_genome_splits.py`)
Stage 1 / 2A ablation (Table 2; Fig. 2a–b)	9 bacterial and archaeal reference genomes × 9 configurations (81 runs)	DIAMOND, Pfam, InterProScan	Stage 1 priors singly and jointly; 2A (`-enforce-stage2`, remove or relax)	Stage 0 noisy-OR	`pillar1_2_ablation_summary.tsv`, `pillar1_2_ablation_per_genome.tsv`	P02
Cross-clade perturbation (Fig. 2c; Table S18)	Same 9 genomes; 100 injected protein–term pairs per genome under two injection designs	DIAMOND, Pfam, InterProScan plus injected claims	Configurations A1 and A6–A9 (2A: `-enforce-stage2`)	Stage 0 noisy-OR	`pillar1_2_perturbation_designs.tsv`, `pillar1_2_perturbation_summary.tsv`	P02
Reference panels (Fig. 2d; Table S2)	13 prokaryotes (11 bacteria, 2 archaea); 21 bacteria	13 genomes: DIAMOND, Pfam, InterProScan; 21 genomes: DIAMOND (Pfam part truncated)	Stage 1 + 2A relax (`-enforce-stage2`)	13 genomes: NCBI PGAP, Pfam, InterProScan, DIAMOND, metagenomic-deepFRI; 21 genomes: ESM2-DeepGOPlus, ProteInfer, CLEAN, ESM-2 t30 k-NN	`pillar1_suite2_panel13_21_sota.tsv`, `pillar1_panel11_structural_audit.tsv`	P03
1,000 RefSeq genomes (Table S8)	1,000 RefSeq bacterial genomes	DIAMOND, Pfam, InterProScan (reused)	Stage 1 + 2A relax (`-enforce-stage2`)	DeepGO-SE, SPROF-GO, DeepGraphGO, DeepFRI, TALE, InterProScan	`pillar1_suite2_1000bacteria_sota.tsv`	P12 (GSPA rows pending)
Multi-kingdom proteomes (Table S9)	88 proteomes: 47 bacteria, 15 archaea, 6 eukaryotes, 20 viruses	DIAMOND (target species and genus excluded), Pfam; InterProScan for 19 genomes	Stage 1 + 2A relax, 1 or 3 iterations (`-enforce-stage2`)	Stage 0 noisy-OR	`multikingdom_summary_by_domain.tsv`, `multikingdom_per_genome_eval.tsv`, `multikingdom_bootstrap_ci.tsv`	P15 (runs: `scripts/run_multikingdom_batch_unimatrix.py`)
Leave-subunit-out (Table 3; Fig. 3a–b; Table S19)	175 trials in 6 prokaryotic genomes; two decoy designs (unrelated loci, within operon)	Attenuated synthetic claims for the held-out subunit and 5 decoys; real DIAMOND and InterProScan channels as comparators	Coherence prior, with and without operon context; 2A (`-enforce-stage2`); Stage 3 through the agent harness	Attenuated input, real DIAMOND and InterProScan channels, operon vote	`pillar3_complex_lso.tsv`, `pillar3_complex_lso_mcnemar.tsv`	P04
Eukaryote anatomy (Table 4; Fig. 3c–d)	6 eukaryotes; 5,546 proteins with real claims; 300 injected cross-clade claims per species (44 for *P. patens*)	Real blind DIAMOND and Pfam claims	Configurations E1–E5: priors, 2A remove or relax (`-enforce-stage2`)	Direct GO taxon rules; deletion versus relaxation	`pillar3_eukaryote_anatomy.tsv`, `pillar3_relax_vs_remove.tsv`, `pillar3_eukaryote_endogenous_vs_injected.tsv`	P05
Leave-reaction-out (Table 5; Figs. 4 and 5c)	*E. coli* K-12: 256 reactions (strict) and 414 reactions under four masking protocols; *B. subtilis* 168: 144; *A. aeolicus* VF5: 101	Track A: homology-masked sequence channels; Track B adds Swiss-Prot ESM-2 reaction centroids (not homology-masked)	Stage 1 with RLGC (`-rlc-suggester`) and coherence	Ranking without target evidence, RLGC alone, RLGC + GBDT, single features	`pillar2_lro15_summary.tsv`, `pillar4_lro_50_cases.jsonl`	P06 (`benchmark/leave_reaction_out/`)

Benchmark (manuscript item)	Cohort	Stage 0 channels	Stages run (options)	Comparators	Result files	Pipeline
Stoichiometric audit (Fig. 5a; Table S12a)	108 bacterial genomes; DeepProZyme draft at 0.50	Not applicable	Stage 2B (Python refiner)	Raw draft, METEOR v8 (archived solutions), Grow-Support repair	`pillar2_invariant_audit_summary.tsv`, `pillar2_invariant_audit_108.tsv`	P07
Price-149 (Table 6; Fig. 5b; Table S12c)	148 RB-TnSeq enzymes from 22 hosts	CLEAN, MAPred, DeepProZyme, EnzBERT, GraphEC, TopEC, singly and pooled	Stage 2B probe and RLGC, Stage 1 neighbour coherence (evaluation script)	Six standalone predictors; METEOR v8 (archived repairs)	`pillar2_price149_all6_real.tsv`, `pillar2_price149_by_stratum.tsv`, `pillar2_price149_per_protein_real.tsv`	P08
BacDive phenotypes (Table 7; Fig. 5d; Table S13)	2,332 carbon-source assays (791 strains); 17,496 enzyme assays (3,164 strains); 50,075 strain–EC pairs (10 ECs)	Binary gapseq, ModelSEED, and CarveMe calls	Track 1 and Track 2 (evaluation script)	Re-implemented L_1-SVM, random forest, GBDT	`pillar3_bacdive_track1_summary.tsv`, `pillar3_bacdive_track2_summary.tsv`, `pillar3_bacdive_per_fold_cv.tsv`	P09 (`benchmark/unified/pillar3_bacdive/`)
Stage 3 benchmark v2 (Table 8; Fig. 6; Tables S14, S20, S21)	80 anomalies from 7 genomes; LRO-50 (50 cases); trigger rates on the 88 multi-kingdom proteomes	Bacteria: DIAMOND, Pfam, InterProScan; eukaryotes: DIAMOND, Pfam	Stage 1 + 2A relax (`-enforce-stage2`, dark-matter suggester); Stage 3 through the agent harness	Stage 1 + 2 only, posterior threshold, LR / RF / GBDT / MLP, single features, agent without guardrail	`pillar4_v2_*.tsv`, `pillar4_v2_transcripts_*.jsonl`	P10
AllTheBacteria (Section S7; Tables S16 and S17)	72,533 genomes (alignment survey); 7,009 and 4,592 genomes (integrator)	GSPA-Seq channels merged into one source per claim (no ESM-2 forward pass, no `hmmsearch`)	Stage 1 + 2A relax (`-enforce-stage2`)	Stage 0 noisy-OR	`allthebacteria/insight3b_phylum_summary.tsv`, `allthebacteria/atb_java_summary*.json`	P11 (Stage D running)
Biosynthetic gene clusters (Table S10)	antiSMASH-annotated genomes	DIAMOND, Pfam, InterProScan	Stage 1 + 2A relax	Stage 0, operon prior only	`pillar2_3_antismash_bgc_specificity_v2.tsv`	P13
MAG pathway gaps (Table S11)	100 high-quality Empty Quarter MAGs (88 bacterial, 12 archaeal; 17 phyla)	DIAMOND (experimental-GO Swiss-Prot), Pfam (`hmmsearch`); no InterProScan	Stage 1 + 2A relax with gapseq gaps, operons, and RLGC (`-enforce-stage2`, `-rlc-suggester`)	Stage 0 noisy-OR; gapseq 1.4.0	`pillar3_dark_matter_mags_v2.tsv`, `pillar3_dark_matter_mags_v2_per_mag.tsv`	P14
EC holdout and 95-genome EC masking (Tables S5 and S6)	125 post-cutoff Swiss-Prot enzymes; 95 bacterial genomes under four masking tiers	CLEAN, DeepProZyme, EnzBERT	Stage 2B single-predictor and three-predictor acceptance (evaluation script)	Standalone predictors, METEOR v8	`scenario1_swissprot_ec_holdout125_real.tsv`, `scenario1_95genomes_ec_homology_real.tsv`	none (`scripts/eval_scenario1_holdouts_real.py`, archived inputs)
Isolated-protein CAFA (Table S15)	272 post-t_0 targets	As in the whole-genome CAFA benchmark	Stage 0, Stage 1 + 2A	14 per-protein methods	`suite1_cafa6_prospective_summary.tsv`	none (`scripts/eval_scenario1_holdouts_real.py`, archived inputs)

Supplementary Table S2 reports the ancestor-propagated evaluation of annotation pipelines, single Stage 0 channels, and GSPA on the 13-genome and 21-genome reference panels. Every method was scored by one evaluator from raw outputs on disk; methods that appear in the panel definitions but have no raw output are listed as not evaluated. Against experimental annotations on the 13-genome panel, GSPA reached the highest micro $F_{\max}$, followed by InterProScan, DIAMOND, Pfam, NCBI PGAP, and metagenomic-deepFRI.

S1.1 Stage 0 calibration, reliability weights, and correlation groups

Stage 0 is implemented in `CalibrationTable.groovy`, `EvidenceCombiner.groovy`, and `EvidenceType.groovy` (package `gspa.integration`). A claim record that carries a calibrated probability is used as given; otherwise its raw score $s(c)$ is mapped to $\tilde{q}(c) = \mathrm{clip}(\sigma_{\log}(a_e s(c) + b_e), 10^{-4}, 1 - 10^{-4})$ with source-specific coefficients (a_e, b_e): $(6.0, -3.0)$ for DIAMOND and MMseqs2 (raw score = percent identity / 100), $(4.0, -1.5)$ for HMMER and Pfam, $(5.0, -2.0)$ for InterProScan and FoldSeek, $(3.5, -1.3)$ for eggNOG-mapper, $(3.0, -1.0)$ for SignalP and DeepTMHMM, $(3.0, -1.2)$ for operon claims, $(4.0, -1.8)$ for gapseq, $(8.0, -4.0)$ for AMRFinder, $(5.0, -2.0)$ for dbCAN, $(4.0, -1.5)$ for antiSMASH, and $(4.0, -2.0)$ for every other source, which includes the deep-learning and protein-language-model predictors.

Table S2. Ancestor-propagated evaluation on the 13-genome and 21-genome reference panels. The 13-genome panel holds 13 prokaryotes (11 bacteria and 2 archaea) and the 21-genome panel 21 bacteria. Truth and predictions are propagated over `is_a` and `part_of`, and the three GO roots are removed. Experimental-truth metrics ("exp."; 323 proteins on the 13-genome panel and 228 on the 21-genome panel) are pooled over genomes; all-evidence metrics ("all") are means of per-genome values. About 85% of the all-evidence truth derives from electronic annotation, much of it assigned by InterPro, which favours InterProScan and every configuration with an InterProScan channel. EC $F_{\max}$ (21-genome panel only) uses exact four-digit matches against the Swiss-Prot EC numbers of the panel proteins and is pooled. On the 13-genome panel GSPA integrates DIAMOND, Pfam, and InterProScan claims with all Stage 1 priors and Stage 2A relaxation. No InterProScan output exists for the 21-genome panel and its Pfam claims hold about 15 GO pairs per genome, so the GSPA row there is effectively DIAMOND-only. The ESM-2 k-NN row uses the 150M-parameter t30 model with the other genomes of a 30-genome panel as references. Methods without raw prediction output are listed as not evaluated. Source: `results/pillar1_suite2_panel13_21_sota.tsv` (pipeline P03).

Panel	Method	Genomes	Micro $F_{\max}$ exp.	CAFA $F_{\max}$ exp.	$S_{\min}$ exp.	Micro $F_{\max}$ all	CAFA $F_{\max}$ all	$S_{\min}$ all	EC micro $F_{\max}$	EC CAFA $F_{\max}$
13 genomes	NCBI PGAP (Li et al. 2021)	13	0.2123	0.2141	45.81	0.6003	0.5825	58.68	–	–
	HMMER 3.4 / Pfam-A + pfam2go	13	0.2401	0.2404	41.02	0.5743	0.5865	66.68	–	–
	InterProScan 5.72-103.0	13	0.3276	0.3220	41.32	0.8843	0.8694	21.79	–	–
	DIAMOND-BLASTp	13	0.3030	0.3153	43.12	0.7712	0.7595	31.47	–	–
	metagenomic-deepFRI (Bezshapkin et al. 2026)	13	0.2008	0.1867	42.61	0.5211	0.4626	70.54	–	–
	GSPA Stage 1+2A (3-Channel D+P+IPR)	13	0.3498	0.3442	42.40	0.9216	0.9037	12.02	–	–
21 genomes	ESM2-DeepGOPlus (2024)	21	0.1538	0.1718	53.20	0.2222	0.2730	97.98	–	–
	ProteInfer (Sanderson et al. 2023)	21	0.4028	0.3999	61.35	0.6745	0.6409	47.89	0.3774	0.3749
	CLEAN Contrastive EC (Yu et al. 2023)	21	–	–	–	–	–	–	0.9393	0.9492
	DIAMOND-BLASTp	21	0.2942	0.3332	52.11	0.7438	0.7430	34.30	–	–
	GSPA Stage 1+2A (DIAMOND-only Stage 0)	21	0.3167	0.3456	51.13	0.7643	0.7509	32.94	–	–
	ESM-2 t30 150M kNN (k=10, leave-one-genome-out over 30-genome panel)	21	0.3363	0.3487	53.22	0.8507	0.8312	24.37	–	–
	HMMER 3.4 / Pfam-A + pfam2go	*not evaluated (no raw prediction output)*								
	InterProScan 5.72-103.0	*not evaluated (no raw prediction output)*								
	FoldSeek + ProstT5 3Di (2024)	*not evaluated (no raw prediction output)*								
	ProtT5 kNN (Elnaggar et al. 2021)	*not evaluated (no raw prediction output)*								
	ESM-2 650M kNN (Lin et al. 2023)	*not evaluated (no raw prediction output)*								
	GSPA Stage 1+2 (3-Channel Core)	*not evaluated (no raw prediction output)*								
	GSPA Stage 1+2 (7-Channel All-Pooled)	*not evaluated (no raw prediction output)*								
9 genomes	gLM / gLM2 Genomic LM (Hwang et al. 2024)	*not evaluated (no raw prediction output)*								

Each claim is weighted by the reliability r_e of its evidence type (method `defaultReliability` of `EvidenceCombiner`): 0.70 for sequence similarity and orthology, 0.75 for sequence domains, structure-based deep learning, and BGC and virulence-factor predictors, 0.80 for structural similarity and CAZyme predictors, 0.85 for AMR predictors, 0.65 for sequence deep learning and protein-language-model embeddings, 0.55 for metabolic context and genomic language models, 0.50 for localization and sequence motifs, 0.45 for genomic context, and 0.60 for the remaining types (dark-matter, reaction-local-context, cross-genome, and ranker claims, and region-level predictors). A `-theta` file can override these weights; the per-benchmark run scripts list the files they pass.

For each protein–function pair, claims are grouped by the correlation group of their evidence type (method `correlationGroup` of `EvidenceType`). There are ten shared groups (homology, protein-sequence machine learning, structure, context, genomic language models, inferred context, motif, localization, region features, and viral) and one group for each of the four domain-specific predictor types (AMR, CAZyme, BGC, and virulence factor). The homology group contains sequence similarity, sequence domain, and orthology claims, so DIAMOND, Pfam, and InterProScan claims for the same pair count once, through the strongest of them. Within each group only the claim with the largest $r_e\tilde{q}(c)$ contributes (the second-opinion bonus is 0 by default), its probability is clipped to $[0.001, 0.999]$, the group probabilities are combined by noisy-OR, and the result is converted to log-odds clipped to $[-12, 12]$.

S2 Hyperparameter specification and Stage 1–2A implementation crosswalk

Supplementary Table S3 lists the 18 configurable hyperparameters of Stage 1 and Stage 2B with their configuration keys, source modules, roles, and default values (penalties are stored as positive magnitudes and applied with negative sign in log-odds space). Stage 1 keys are read by `IntegrateCommand.groovy` from a `-theta` file and consumed by `IterativeRefiner.groovy` and the prior modules; the fixed candidate-partner gate 0.08 (`CoherencePrior.groovy`), the co-operonic complex partner weight 0.95 (`GenomicContextPrior.groovy`), the Stage 2A ancestor relaxation factor 1.0 (`ConsistencyEnforcer.groovy`), and the obligate-complex singleton prune threshold 0.45 (`IntegrateCommand.groovy`) are constants. Stage 2B parameters live in `stoichiometric_locus_refiner.py`, together with the fixed elemental-imbalance penalty $w_{\text{imb}} = 100.0$. The defaults were set during development on *Escherichia coli* K-12 MG1655 and were applied unchanged to every genome within each reported experiment. Two of them changed during the study: the integrator build of 2026-10-17 (v2.1.0) lowered α_{cons} from 3.0 and α_{anat} from 3.5 to 0.68. Every benchmark run with the v2.1.0 jar uses 0.68; Supplementary Table S18 reports the perturbation experiment under both builds.

Table S3. The 18 configurable GSPA v2.1.0 hyperparameters of Stages 1 and 2B.

Stage / module	Symbol	Parameter key	Source module	Role in inference	Default
Stage 1 (B_1)	α_{ess}	`alpha_ess`	`EssentialityPrior.groovy`	Scale of the essential-function rescue	1.50
Stage 1 (B_2)	α_{coh}	`alpha_coh`	`CoherencePrior.groovy`	Scale of process, pathway, and complex coherence	1.00
Stage 1 (B_2)	w_{part}	`alpha_complex_boost`	`CoherencePrior.groovy`	Obligate heteromeric complex partner weight ($0.65\, w_{\text{part}} = 0.5525$ when $N_{\text{strong}} \geq 2$)	0.85
Stage 1 (B_2)	w_{sing}	`alpha_complex_penalty`	`CoherencePrior.groovy`	Isolated obligate-complex singleton penalty magnitude	0.75
Stage 1 (B_3)	α_{cons}	`alpha_cons`	`ConsistencyPrior.groovy`	SAT4J taxonomic conflict penalty magnitude	0.68
Stage 1 (B_4)	α_{anat}	`alpha_anat_penalty`	`AnatomicalPrior.groovy`	Taxon-forbidden Uberon/PO/CL structure penalty magnitude	0.68
Stage 1 (B_4)	α_{boost}	`alpha_anat_boost`	`AnatomicalPrior.groovy`	Active-structure developmental boost	0.65
Stage 1 (B_5)	α_{gap}	`alpha_gap`	`GapFillingPrior.groovy`	Metabolic gap-filling scale	1.50
Stage 1 (B_6)	α_{ctx}	`alpha_ctx`	`GenomicContextPrior.groovy`	Co-operonic context scale	1.00
Stage 1 (B_6)	$\alpha_{\text{gap_ctx}}$	`alpha_gap_ctx`	`GenomicContextPrior.groovy`	Co-operonic metabolic gap multiplier on $\alpha_{\text{ctx}}\, c_O(W)$	1.50
Stage 1 (iteration)	γ_{iter}	`damping`	`IterativeRefiner.groovy`	Log-odds damping weight	0.50
Stage 1 (iteration)	$K_{\max}$	`max_iter`	`IterativeRefiner.groovy`	Maximum Stage 1 iterations	6
Stage 1 (iteration)	ϵ	`epsilon`	`IterativeRefiner.groovy`	Probability convergence tolerance	0.005
Stage 2B (SLR)	τ_{core}	`tau_core`	`stoichiometric_locus_refiner.py`	Core reaction probability threshold $S_{\text{rxn}}(j)$	0.50
Stage 2B (SLR)	α_{bio}	`alpha_bio`	`stoichiometric_locus_refiner.py`	Minimum fraction of $v_{\text{bio}}^{\max}$ in pFBA	0.10
Stage 2B (SLR)	η_{null}	`tau_null`	`stoichiometric_locus_refiner.py`	Prior mass of the unassigned-gene hypothesis $h_{\varnothing}$	0.05
Stage 2B (SLR)	γ_{FBA}	`gamma_fba`	`stoichiometric_locus_refiner.py`	Backward locus responsibility log-odds scale	2.20
Stage 2B (SLR)	γ_{damp}	`gamma_damp`	`stoichiometric_locus_refiner.py`	Geometric damping across outer passes	0.60

S2.1 Targeted rescue and specificity terms of the Stage 1 priors

The iterative refiner records the number of supporting claims for every protein–function pair, and when these counts are present (which is the case in every `gspa integrate` run) the essentiality (B_1), coherence (B_2), gap-filling (B_5), and genomic-context (B_6) priors apply a targeted form. Each of these priors collects borderline candidates for every unmet invariant (an uncovered essential category, a missing step of a process or pathway, an open metabolic gap, or a missing pathway term in an operon), ranks them, and raises only the top-ranked candidate. A candidate is a pair whose Stage 0 likelihood probability $P^{(0)}$ lies in $[0.16, 0.50]$ for B_1 and in $[0.18, 0.50]$ for B_2, B_5, and B_6. Candidates are ranked by a rescue tier and then by $P^{(0)}$ (method `candidateRescueTier` of `IntegrationState`): tier 4 if $P^{(0)} \geq 0.45$, tier 3 if $P^{(0)} \geq 0.40$, tier 2 if the protein's highest GO likelihood is below 0.50, tier 1 if the pair has at least two supporting claims, and tier 0 otherwise. B_2, B_5, and B_6 raise the top candidate only when its tier is at least 1; B_1 has no tier requirement. With $L^{(0)}$ the Stage 0 log-odds of the selected pair, the boosts are

$$
\begin{aligned}
B_1 &: \max\bigl(\min(2.0\,\alpha_{\text{ess}}, 1.30\,\alpha_{\text{ess}}),\ 0.14 - L^{(0)}\bigr) = \max\bigl(1.30\,\alpha_{\text{ess}},\ 0.14 - L^{(0)}\bigr),\\
B_2 &: \max\bigl(1.80\,\alpha_{\text{coh}}\, w,\ 0.14 - L^{(0)}\bigr),\\
B_5 &: \max\bigl(1.15\,\mu,\ 0.14 - L^{(0)}\bigr),\ \mu = \alpha_{\text{gap}}\ (0.7\,\alpha_{\text{gap}} \text{ for gapseq-guessed gaps}),\\
B_6 &: \max\bigl(1.65\,\alpha_{\text{ctx}},\ 0.14 - L^{(0)}\bigr),
\end{aligned}
$$

where w is the process or pathway weight defined below (source lines: `EssentialityPrior.groovy` 106–124, `CoherencePrior.groovy` 186–222, `GapFillingPrior.groovy` 114–125, and `GenomicContextPrior.groovy` 206–217). The second argument guarantees that the boost alone lifts the pair to log-odds 0.14 (probability 0.535) before damping. In this targeted form the uniform base terms written in the Methods for B_1, the process and pathway part of B_2, B_5, and the pathway-consensus part of B_6 are not applied to the other candidates; the obligate-complex weights of B_2 and the co-operonic complex partner weight of B_6 are applied as defined in the Methods. When claim counts are absent (programmatic use of the priors without the refiner), the priors fall back to the base terms.

The same four priors also attenuate uncorroborated secondary terms (method `computeProteinSpecificityPrior` of `IntegrationState`). On a protein whose top GO term in an aspect has $P^{(0)} \geq 0.55$ and at least two supporting claims, every other term of that aspect with a single supporting claim and $P^{(0)} \in [0.40, 0.46)$ that lies outside the prior's own domain terms receives the adjustment $\max(-\kappa, -0.52 - L^{(0)})$, which moves it towards log-odds -0.52 (probability 0.373). The magnitudes are $\kappa = 0.30\,\alpha_{\text{ess}}$ (B_1), $0.42\,\alpha_{\text{coh}}$ (B_2), $0.35\,\alpha_{\text{gap}}$ (B_5), and $0.38\,\alpha_{\text{ctx}}$ (B_6, restricted to proteins in operons). A pair is adjusted at most once per iteration, by the first prior that reaches it.

S2.2 Process and pathway weights of the coherence prior

The coherence prior computes two weights for missing terms (`CoherencePrior.groovy`:91–123). For GO `has_part` process dependencies triggered by the baseline annotation, the weight is $w_{\text{proc}} = \max(0.25, n_{\text{sat}}/n_{\text{trig}})$, where n_{trig} and n_{sat} are the numbers of triggered and satisfied dependencies, so the weight grows as the genome's process coverage grows. For every pathway or module of the pathway database (KEGG pathways and modules) with at least one required GO term annotated, the weight of its missing required terms is $w_{\text{path}} = \max(0.35, 0.35 + 0.65\,c_W)$ with step coverage c_W; missing terms of pathways with $c_W < 0.15$ do not enter the targeted-rescue set. Each missing term takes the largest weight over the processes and pathways that require it, and w in the B_2 boost above is this weight.

S2.3 Stages and options used in each benchmark

Stage 2A runs only when `-enforce-stage2` is given, which is the case for the ablation configurations A8 and A9, the perturbation configurations A8 and A9, the reference panels, the 1,000-genome and multi-kingdom runs, the leave-subunit-out Stage 2 configuration, the eukaryote configurations E2 and E4 (remove) and E5 (relax), and the Stage 3 benchmark runs. The reaction-local genomic context suggester (`-rlc-suggester`) is used in the leave-reaction-out runs (`benchmark/leave_reaction_out/run_ablation_rlgc*.py`) and in the MAG runs, and the dark-matter suggester in the Stage 3 benchmark runs whenever a gap table is present. Stage 2B and Stage 3 are never run through `-enable-fba-refiner` or `-enable-agent`; the benchmarks that report them call the Python refiner and the agent harness directly (Supplementary Table S1).

S2.4 Whole-genome CAFA splits, enzyme holdouts, and 95-genome homology masking

The whole-genome CAFA benchmark (Supplementary Table S4) withholds six complete prokaryotic genomes (20,697 coding sequences). The training snapshot t_0 is the `train_terms.tsv` file of the CAFA6 Kaggle release dated 2025-12-10 (82,404 annotated proteins in the rollover split). Targets and their terms come from the `truth_dual` files of the six genomes, which hold GOA annotations with experimental evidence codes; the GOA release from which these files were built is not recorded. The 1,060 targets of the two time splits are proteins of the six genomes with experimental annotations in the truth files and none in the t_0 snapshot, and the homology splits evaluate all 3,315 proteins with experimental annotations. On every split, the per-protein baselines are retrained on that split's training set. DeepFRI, DeepGraphGO, TALE+, SPROF-GO, DeepGO-SE, and DPFunc are re-implementations that take ESM-2 3B embeddings as input: DeepFRI uses sequence-band smoothing in place of contact maps, DeepGraphGO uses a DIAMOND similarity graph, SPROF-GO reshapes the mean embedding into 16 pseudo-tokens, and sequences are truncated to 256 residues. The official pretrained DeepFRI and DeepGO-SE models are evaluated on the two time splits.

Stage 0 in this benchmark pools six retrained channels (DIAMOND k-NN, InterPro-LR, DeepGOCNN, and ESM-2 3B k-NN, MLP, and C-HMCNN). For each split, each channel's reliability per aspect is calibrated on 600 pre-t_0 validation proteins outside the six genomes: the best protein-centric F over thresholds $\{0.15, 0.25, 0.35, 0.45\}$ (at least 0.10) is raised to the power 1.35, multiplied by 0.48 for the three ESM-2 channels, and clipped to $[0.05, 0.88]$. The DeepGOCNN, ESM-2 MLP, and ESM-2 C-HMCNN reliabilities for MF and BP are further multiplied by a factor $h \in \{1.0, 0.75, 0.55, 0.38\}$ chosen on the same validation proteins by the Stage 0 f_w (ties broken towards the larger value). The selected factor was 0.75 on the two time splits and the $\leq 100\%$ homology split and 1.0 on the $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ splits (column h of Supplementary Table S4). Stage 1, 2A, and 2B are implemented in the benchmark script over these channels, with the operons and gapseq gaps of the six genomes. The table also reports violations at a fixed threshold of 0.50 ($V_{0.50}$) next to the count at $\min(\tau^*, 0.25)$.

Table S4. Whole-genome CAFA evaluation on all six splits. Six complete prokaryotic genomes (20,697 coding sequences) are withheld and annotated as whole chromosomes. Targets are proteins with experimental GOA annotations in the `truth_dual` files; the time splits evaluate the 1,060 targets that gained their first experimental annotation after t_0, and the homology splits evaluate all 3,315 targets. f_w is the information-accretion-weighted $F_{\max}$ (CAFA IA weights), $F_{\max}$ the protein-centric $F_{\max}$, and $S_{\min}$ the minimum semantic distance, each averaged over MF, BP, and CC. V counts propagated protein–term pairs with a taxon-forbidden term scored at or above $\min(\tau^*, 0.25)$, where τ^* is the method's f_w-optimal threshold for the aspect; $V_{0.50}$ counts them at a fixed threshold of 0.50. h is the down-weighting factor of the DeepGOCNN, ESM-2 MLP, and ESM-2 C-HMCNN channels (MF and BP) inside the Stage 0 pool of the GSPA rows, selected on each split's 600 validation proteins from $\{1.0, 0.75, 0.55, 0.38\}$. The official pretrained DeepFRI and DeepGO-SE models are evaluated on the two time splits only. Source: `results/cafa_whole_genome_time_and_homology_splits.tsv` (pipeline P01, `scripts/run_cafa_whole_genome_splits.py`).

Method	f_w MF	f_w BP	f_w CC	f_w mean	$F_{\max}$ mean	$S_{\min}$ mean	V	$V_{0.50}$	h
Time split, pre-t_0 annotations of the six genomes kept in training (rollover): 82,404 training proteins, 1,060 evaluation targets									
Naive Term-Frequency Prior (CAFA Baseline 1)	0.3419	0.1643	0.3752	0.2938	0.3349	11.73	1,570	0	–
DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4829	0.3361	0.3070	0.3753	0.4064	10.46	988	201	–
InterPro Domain Logistic Regression (InterPro-LR)	0.5622	0.4131	0.3680	0.4477	0.5017	9.70	916	214	–
DeepGOCNN (1D-CNN Sequence Model, Retrained)	0.3730	0.1797	0.2879	0.2802	0.3520	11.87	656	9	–
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	0.5499	0.3437	0.3463	0.4133	0.4681	10.38	1,026	132	–
ESM-2 Embedding kNN (goPredSim, Retrained)	0.4746	0.3083	0.4583	0.4138	0.4835	10.50	445	97	–
ESM-2 MLP (Flat BCE, Retrained)	0.5002	0.3312	0.4714	0.4343	0.4981	10.40	357	103	–
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	0.5095	0.3362	0.5141	0.4533	0.5186	10.28	237	71	–
DeepFRI (Pretrained Official)	0.2655	0.1705	0.2886	0.2415	0.2879	11.76	1,300	24	–
DeepGO-SE (Pretrained Official)	0.5077	0.3014	0.5491	0.4527	0.4855	10.09	113	33	–
DeepFRI (GCN + LM, Retrained)	0.4408	0.2752	0.4366	0.3842	0.4620	10.89	61	16	–
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5051	0.3353	0.5249	0.4551	0.5162	10.25	79	46	–
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5435	0.3584	0.4624	0.4548	0.5202	10.13	216	32	–
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5771	0.3691	0.5122	0.4861	0.5467	9.80	99	31	–
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.5068	0.2799	0.4944	0.4270	0.4984	10.67	83	31	–
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5143	0.3476	0.5279	0.4633	0.5228	10.10	318	34	–
GSPA Stage 0 (Per-Protein Noisy-OR)	0.5816	0.4005	0.4937	0.4919	0.5392	9.62	935	251	0.75
GSPA-Seq (Ours, Per-Protein Integrator)	0.5808	0.3980	0.5059	0.4949	0.5469	9.62	938	318	0.75
GSPA + Operon Genomic Context Only (B6)	0.5820	0.3980	0.5108	0.4969	0.5467	9.61	938	318	0.75
GSPA + Pathway & Complex Coherence Only (B1/B2)	0.5811	0.3977	0.5054	0.4947	0.5468	9.62	938	318	0.75
GSPA + Metabolic Gap-Filling & FBA Only (B5 + Stage 2B)	0.5808	0.3980	0.5059	0.4949	0.5469	9.62	938	318	0.75
GSPA + Taxon & Anatomical Constraints Only (B3/B4 + Stage 2A)	0.5805	0.3981	0.5206	0.4997	0.5486	9.57	0	0	0.75
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.5810	0.3987	0.5262	0.5020	0.5485	9.56	0	0	0.75
Time split, six genomes withheld from training: 79,880 training proteins, 1,060 evaluation targets									
Naive Term-Frequency Prior (CAFA Baseline 1)	0.3419	0.1643	0.3276	0.2779	0.3349	11.89	1,570	0	–
DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4581	0.3022	0.2345	0.3316	0.3690	10.70	1,195	229	–
InterPro Domain Logistic Regression (InterPro-LR)	0.5489	0.4084	0.3383	0.4319	0.4852	9.80	1,114	229	–
DeepGOCNN (1D-CNN Sequence Model, Retrained)	0.3704	0.1613	0.2225	0.2514	0.3357	11.95	1,406	38	–
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	0.5237	0.3127	0.2528	0.3630	0.4361	10.66	1,532	155	–
ESM-2 Embedding kNN (goPredSim, Retrained)	0.4304	0.2738	0.2249	0.3097	0.3939	11.14	864	142	–
ESM-2 MLP (Flat BCE, Retrained)	0.4809	0.2846	0.3531	0.3729	0.4538	10.68	620	173	–
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	0.4949	0.3035	0.3812	0.3932	0.4655	10.69	572	173	–
DeepFRI (Pretrained Official)	0.2655	0.1705	0.2886	0.2415	0.2879	11.76	1,300	24	–
DeepGO-SE (Pretrained Official)	0.5077	0.3014	0.5491	0.4527	0.4855	10.09	113	33	–
DeepFRI (GCN + LM, Retrained)	0.4257	0.2592	0.2974	0.3274	0.4216	11.22	308	20	–
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5080	0.3145	0.3716	0.3980	0.4753	10.34	200	51	–
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5411	0.3380	0.3085	0.3959	0.4744	10.47	371	47	–
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5272	0.3378	0.3277	0.3976	0.4746	10.35	655	32	–
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4789	0.2588	0.3560	0.3646	0.4413	10.99	257	33	–
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5034	0.3117	0.3599	0.3917	0.4666	10.52	631	40	–
GSPA Stage 0 (Per-Protein Noisy-OR)	0.5693	0.3789	0.3560	0.4347	0.4914	10.03	1,392	310	0.75
GSPA-Seq (Ours, Per-Protein Integrator)	0.5687	0.3807	0.3588	0.4361	0.4929	10.08	1,426	450	0.75
GSPA + Operon Genomic Context Only (B6)	0.5658	0.3786	0.3646	0.4363	0.4919	10.09	1,426	450	0.75
GSPA + Pathway & Complex Coherence Only (B1/B2)	0.5698	0.3805	0.3586	0.4363	0.4932	10.07	1,426	450	0.75
GSPA + Metabolic Gap-Filling & FBA Only (B5 + Stage 2B)	0.5689	0.3807	0.3588	0.4361	0.4929	10.08	1,426	450	0.75
GSPA + Taxon & Anatomical Constraints Only (B3/B4 + Stage 2A)	0.5684	0.3808	0.3924	0.4472	0.4938	9.99	0	0	0.75
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.5657	0.3795	0.3927	0.4460	0.4934	9.99	0	0	0.75

Method	f_w MF	f_w BP	f_w CC	f_w mean	$F_{\max}$ mean	$S_{\min}$ mean	V	$V_{0.50}$	h
Homology split, six species purged ($\leq$ 100% identity): 79,880 training proteins, 3,315 evaluation targets									
Naive Term-Frequency Prior (CAFA Baseline 1)	0.3734	0.1753	0.3890	0.3126	0.3537	12.11	5,274	0	–
DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4922	0.3505	0.2139	0.3522	0.3937	10.44	4,340	620	–
InterPro Domain Logistic Regression (InterPro-LR)	0.5865	0.4802	0.2785	0.4484	0.5072	9.46	5,168	1,000	–
DeepGOCNN (1D-CNN Sequence Model, Retrained)	0.3708	0.1987	0.2389	0.2695	0.3533	12.12	4,127	123	–
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	0.5461	0.3593	0.2509	0.3854	0.4572	10.40	1,868	355	–
ESM-2 Embedding kNN (goPredSim, Retrained)	0.4401	0.3117	0.2339	0.3286	0.4080	11.08	4,289	372	–
ESM-2 MLP (Flat BCE, Retrained)	0.5039	0.3173	0.3259	0.3824	0.4646	10.79	1,251	227	–
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	0.5133	0.3287	0.3457	0.3959	0.4727	10.74	1,517	208	–
DeepFRI (GCN + LM, Retrained)	0.4214	0.2951	0.2951	0.3372	0.4279	11.41	843	29	–
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5270	0.3508	0.3321	0.4033	0.4847	10.45	373	67	–
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5684	0.3865	0.3156	0.4235	0.4929	10.29	1,100	125	–
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5501	0.3878	0.3009	0.4129	0.4929	10.18	823	38	–
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4926	0.3020	0.3128	0.3691	0.4525	11.04	730	40	–
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5264	0.3460	0.3105	0.3943	0.4754	10.51	1,982	57	–
GSPA Stage 0 (Per-Protein Noisy-OR)	0.6008	0.4382	0.3262	0.4551	0.5170	9.81	4,865	930	0.75
GSPA-Seq (Ours, Per-Protein Integrator)	0.6018	0.4381	0.3248	0.4549	0.5189	9.82	4,716	1,239	0.75
GSPA + Operon Genomic Context Only (B6)	0.6000	0.4372	0.3335	0.4569	0.5184	9.81	4,716	1,239	0.75
GSPA + Pathway & Complex Coherence Only (B1/B2)	0.6086	0.4386	0.3247	0.4573	0.5206	9.79	4,716	1,239	0.75
GSPA + Metabolic Gap-Filling & FBA Only (B5 + Stage 2B)	0.6021	0.4381	0.3248	0.4550	0.5190	9.82	4,716	1,239	0.75
GSPA + Taxon & Anatomical Constraints Only (B3/B4 + Stage 2A)	0.6017	0.4384	0.3550	0.4651	0.5198	9.78	0	0	0.75
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6059	0.4373	0.3627	0.4686	0.5206	9.75	0	0	0.75
Homology split, training homologs above 70% identity purged: 79,457 training proteins, 3,315 evaluation targets									
Naive Term-Frequency Prior (CAFA Baseline 1)	0.3734	0.1753	0.3890	0.3126	0.3537	12.11	5,274	0	–
DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4842	0.3473	0.2046	0.3454	0.3878	10.49	4,355	690	–
InterPro Domain Logistic Regression (InterPro-LR)	0.5849	0.4816	0.2771	0.4479	0.5059	9.44	5,206	1,007	–
DeepGOCNN (1D-CNN Sequence Model, Retrained)	0.3749	0.2007	0.2160	0.2639	0.3539	12.14	3,876	116	–
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	0.5507	0.3564	0.2472	0.3847	0.4584	10.45	3,017	314	–
ESM-2 Embedding kNN (goPredSim, Retrained)	0.3128	0.1435	0.2215	0.2259	0.3097	12.59	5,398	456	–
ESM-2 MLP (Flat BCE, Retrained)	0.4975	0.3190	0.3119	0.3761	0.4631	10.79	887	147	–
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	0.5172	0.3254	0.3487	0.3971	0.4804	10.71	1,061	147	–
DeepFRI (GCN + LM, Retrained)	0.4098	0.2830	0.2686	0.3205	0.4085	11.49	2,074	53	–
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5189	0.3498	0.3381	0.4023	0.4827	10.49	514	53	–
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5662	0.3867	0.3223	0.4251	0.4942	10.26	1,875	109	–
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5557	0.3869	0.3294	0.4240	0.4982	10.11	1,858	30	–
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4931	0.2998	0.3092	0.3674	0.4519	11.05	558	23	–
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5200	0.3443	0.3086	0.3910	0.4703	10.53	1,208	54	–
GSPA Stage 0 (Per-Protein Noisy-OR)	0.6033	0.4386	0.3316	0.4578	0.5175	9.80	4,376	734	1.00
GSPA-Seq (Ours, Per-Protein Integrator)	0.6030	0.4379	0.3389	0.4599	0.5225	9.77	4,412	1,072	1.00
GSPA + Operon Genomic Context Only (B6)	0.6042	0.4337	0.3486	0.4622	0.5218	9.77	4,412	1,072	1.00
GSPA + Pathway & Complex Coherence Only (B1/B2)	0.6132	0.4385	0.3381	0.4633	0.5250	9.72	4,412	1,072	1.00
GSPA + Metabolic Gap-Filling & FBA Only (B5 + Stage 2B)	0.6039	0.4379	0.3389	0.4602	0.5226	9.77	4,412	1,072	1.00
GSPA + Taxon & Anatomical Constraints Only (B3/B4 + Stage 2A)	0.6030	0.4383	0.3679	0.4697	0.5234	9.74	0	0	1.00
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6114	0.4335	0.3769	0.4740	0.5244	9.70	0	0	1.00
Homology split, training homologs above 50% identity purged: 76,405 training proteins, 3,315 evaluation targets									
Naive Term-Frequency Prior (CAFA Baseline 1)	0.3734	0.1753	0.3757	0.3081	0.3537	12.11	5,274	0	–
DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4327	0.2930	0.1747	0.3001	0.3455	10.97	6,231	821	–
InterPro Domain Logistic Regression (InterPro-LR)	0.5862	0.4778	0.2764	0.4468	0.5041	9.43	4,919	931	–
DeepGOCNN (1D-CNN Sequence Model, Retrained)	0.3683	0.1946	0.2222	0.2617	0.3499	12.20	3,642	106	–
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	0.5153	0.3206	0.2363	0.3574	0.4341	10.87	2,512	251	–
ESM-2 Embedding kNN (goPredSim, Retrained)	0.3060	0.1366	0.2821	0.2416	0.3148	12.57	6,818	332	–
ESM-2 MLP (Flat BCE, Retrained)	0.4911	0.3080	0.3143	0.3711	0.4563	10.94	1,409	230	–
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	0.5028	0.3113	0.3069	0.3737	0.4617	10.98	1,443	144	–
DeepFRI (GCN + LM, Retrained)	0.4208	0.2792	0.2976	0.3325	0.4125	11.51	2,475	46	–
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5012	0.3290	0.3169	0.3824	0.4633	10.73	446	47	–
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5513	0.3523	0.3069	0.4035	0.4792	10.69	1,761	160	–
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5419	0.3567	0.3144	0.4043	0.4821	10.45	1,458	31	–
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4843	0.2882	0.3149	0.3625	0.4479	11.17	922	21	–
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5041	0.3187	0.3431	0.3886	0.4682	10.80	2,784	71	–
GSPA Stage 0 (Per-Protein Noisy-OR)	0.5955	0.4247	0.3308	0.4504	0.5080	10.06	4,222	747	1.00

Method	f_w MF	f_w BP	f_w CC	f_w mean	$F_{\max}$ mean	$S_{\min}$ mean	V	$V_{0.50}$	h
GSPA-Seq (Ours, Per-Protein Integrator)	0.5962	0.4225	0.3411	0.4533	0.5107	9.99	4,458	1,193	1.00
GSPA + Operon Genomic Context Only (B6)	0.6016	0.4193	0.3550	0.4586	0.5113	9.99	4,458	1,193	1.00
GSPA + Pathway & Complex Coherence Only (B1/B2)	0.6129	0.4235	0.3399	0.4588	0.5148	9.95	4,458	1,193	1.00
GSPA + Metabolic Gap-Filling & FBA Only (B5 + Stage 2B)	0.5975	0.4226	0.3411	0.4537	0.5109	9.99	4,458	1,193	1.00
GSPA + Taxon & Anatomical Constraints Only (B3/B4 + Stage 2A)	0.5962	0.4234	0.3746	0.4647	0.5116	9.96	0	0	1.00
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6109	0.4206	0.3860	0.4725	0.5149	9.87	0	0	1.00
Homology split, training homologs above 30% *identity purged*: 61,323 training proteins, 3,315 evaluation targets									
Naive Term-Frequency Prior (CAFA Baseline 1)	0.3734	0.1625	0.3757	0.3039	0.3477	12.14	5,274	0	–
DIAMOND BLAST-kNN (CAFA Baseline 2)	0.1756	0.1026	0.0707	0.1163	0.1367	12.43	5,519	441	–
InterPro Domain Logistic Regression (InterPro-LR)	0.5788	0.4695	0.2681	0.4388	0.4978	9.23	3,330	458	–
DeepGOCNN (1D-CNN Sequence Model, Retrained)	0.3622	0.1701	0.2425	0.2582	0.3398	12.34	4,226	134	–
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	0.4015	0.1922	0.2344	0.2760	0.3556	12.06	3,980	180	–
ESM-2 Embedding kNN (goPredSim, Retrained)	0.3060	0.1389	0.2571	0.2340	0.3137	12.67	7,719	443	–
ESM-2 MLP (Flat BCE, Retrained)	0.4226	0.2597	0.3299	0.3374	0.4194	11.55	1,066	60	–
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	0.4360	0.2534	0.3633	0.3509	0.4191	11.59	1,245	32	–
DeepFRI (GCN + LM, Retrained)	0.3660	0.2348	0.2099	0.2702	0.3681	11.94	1,949	54	–
DeepGraphGO (InterPro + GraphGCN, Retrained)	0.4201	0.2771	0.3162	0.3378	0.4274	11.48	376	17	–
TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.4549	0.2466	0.3309	0.3442	0.4198	11.76	2,787	110	–
SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.4389	0.2625	0.3291	0.3435	0.4215	11.47	1,063	18	–
DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4124	0.2475	0.3246	0.3282	0.4033	11.81	2,139	19	–
DPFunc (Domain Cross-Attn + GCN, Retrained)	0.4235	0.2665	0.3031	0.3311	0.4162	11.51	3,413	34	–
GSPA Stage 0 (Per-Protein Noisy-OR)	0.6152	0.4430	0.3492	0.4691	0.5083	10.29	2,941	266	1.00
GSPA-Seq (Ours, Per-Protein Integrator)	0.6138	0.4412	0.3636	0.4729	0.5104	10.29	3,268	549	1.00
GSPA + Operon Genomic Context Only (B6)	0.6054	0.4361	0.3831	0.4749	0.5092	10.13	3,268	549	1.00
GSPA + Pathway & Complex Coherence Only (B1/B2)	0.6130	0.4414	0.3636	0.4727	0.5104	10.00	3,268	549	1.00
GSPA + Metabolic Gap-Filling & FBA Only (B5 + Stage 2B)	0.6138	0.4413	0.3636	0.4729	0.5104	10.27	3,268	549	1.00
GSPA + Taxon & Anatomical Constraints Only (B3/B4 + Stage 2A)	0.6138	0.4426	0.3923	0.4829	0.5112	10.24	0	0	1.00
GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6067	0.4375	0.4098	0.4847	0.5100	9.91	0	0	1.00

Supplementary Table S5 reports the post-cutoff Swiss-Prot enzyme benchmark (125 proteins, 293 protein–EC pairs), with standalone predictors, single-predictor refinement, and three-predictor pooling, and Supplementary Table S6 reports the 95-genome enzyme evaluation under four donor-database homology-masking tiers, comparing unfiltered METEOR v8 Top-1 changes with GSPA's single-predictor and three-predictor acceptance filters, with discordant counts (b_{01}, b_{10}) and exact two-sided McNemar p-values. Both tables are transcribed from `results/scenario1_swissprot_ec_holdout125_real.tsv` and `results/scenario1_95genomes_ec_homology_real.tsv`, which `scripts/eval_scenario1_holdouts_real.py` computes from archived prediction files.

Table S5. Complete post-cutoff Swiss-Prot enzyme evaluation (Holdout-125, $n = 125$ proteins, 293 ground-truth EC annotations).

Predictor / Refinement Method	Masking Tier	Top-1 Acc. (Hits)	Top-3 Acc. (Hits)	Top-5 Acc. (Hits)	MRR	Micro $F_{\max}$
CLEAN (2023)	Unfiltered	32.8% (41/125)	40.8% (51/125)	41.6% (52/125)	0.3719	0.4360
DeepProZyme (2023)	Unfiltered	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2750	0.2715
DeepProZyme + METEOR v8 (Single-Pred)	Unfiltered	25.6% (32/125)	31.2% (39/125)	31.2% (39/125)	0.2872	0.2804
DeepProZyme + GSPA Single-Pred (SLR + Coh.)	Unfiltered	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2779	0.2703
EnzBERT (2023)	Unfiltered	32.8% (41/125)	37.6% (47/125)	39.2% (49/125)	0.3531	0.3596
Noisy-OR (3-Method Ensemble)	Unfiltered	33.6% (42/125)	40.8% (51/125)	43.2% (54/125)	0.3807	0.4381
GSPA (3-Method, Iter 1)	Unfiltered	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.4035
GSPA (3-Method, Iter 2)	Unfiltered	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.3891
GSPA (3-Method, Iter 3)	Unfiltered	32.0% (40/125)	43.2% (54/125)	44.0% (55/125)	0.3781	0.3830
EnzBERT	$\leq 70\%$ identity	30.4% (38/125)	35.2% (44/125)	35.2% (44/125)	0.3305	0.3204
EnzBERT	$\leq 50\%$ identity	28.0% (35/125)	33.6% (42/125)	33.6% (42/125)	0.3040	0.3163
EnzBERT	$\leq 30\%$ identity	17.6% (22/125)	24.8% (31/125)	24.8% (31/125)	0.2139	0.2212
CLEAN	$\leq 30\%$ identity	24.0% (30/125)	30.4% (38/125)	32.8% (41/125)	0.2741	0.2299
DeepProZyme	$\leq 70\%$ identity	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2713	0.2715
DeepProZyme	$\leq 50\%$ identity	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2720	0.2684
DeepProZyme	$\leq 30\%$ identity	22.4% (28/125)	31.2% (39/125)	32.8% (41/125)	0.2674	0.2650

Table S6. Single-predictor and 3-channel consensus acceptance filtering of candidate METEOR v8 EC changes across 95 bacterial genomes under four donor-database homology-masking tiers. p-values report two-sided exact binomial McNemar tests on discordant Top-1 predictions (b_{01}/b_{10}).

Predictor	Tier	Genomes / Proteins	Base Top-1 (F_{max})	METEOR v8 Top-1 (Δ)	METEOR b_{01}/b_{10} (p)	GSPA Single-Pred Top-1 (Δ)	Single-Pred b_{01}/b_{10} (p)	GSPA Multi-Pred Top-1 (Δ)	Multi-Pred b_{01}/b_{10} (F_{max})
CLEAN	Unfiltered	73/52,814	0.8214 (0.8796)	0.8234 (+108)	184/76 ($< 10^{-6}$)	0.8228 (+77)	99/22 ($< 10^{-6}$)	0.8243 (+155)	162/7 (0.8816)
CLEAN	$\leq 70\%$	95/69,663	0.8267 (0.8836)	0.8289 (+149)	191/42 ($< 10^{-6}$)	0.8283 (+108)	131/23 ($< 10^{-6}$)	0.8289 (+149)	154/5 (0.8851)
CLEAN	$\leq 50\%$	95/69,663	0.8271 (0.8844)	0.8294 (+162)	208/46 ($< 10^{-6}$)	0.8291 (+139)	156/17 ($< 10^{-6}$)	0.8293 (+155)	158/3 (0.8859)
CLEAN	$\leq 30\%$	94/68,681	0.8288 (0.8833)	0.8306 (+122)	156/34 ($< 10^{-6}$)	0.8302 (+93)	107/14 ($< 10^{-6}$)	0.8306 (+126)	132/6 (0.8845)
DeepProZyme	Unfiltered	95/69,666	0.9022 (0.9331)	0.9010 (−85)	103/188 (1.0×10^{-6})	0.9017 (−33)	24/57 (3.2×10^{-4})	0.9027 (+36)	51/15 (0.9335)
DeepProZyme	$\leq 70\%$	95/69,666	0.8965 (0.9290)	0.8961 (−27)	151/178 (0.152)	0.8963 (−15)	44/59 (0.167)	0.8975 (+67)	78/11 (0.9296)
DeepProZyme	$\leq 50\%$	95/69,666	0.8966 (0.9288)	0.8966 (+1)	163/162 (1.000)	0.8966 (−3)	49/52 (0.842)	0.8979 (+87)	93/6 (0.9297)
DeepProZyme	$\leq 30\%$	95/69,666	0.8967 (0.9289)	0.8970 (+19)	170/151 (0.315)	0.8966 (−4)	51/55 (0.771)	0.8976 (+67)	76/9 (0.9296)
EnzBERT	Unfiltered	95/69,666	0.8430 (0.8707)	0.8416 (−92)	100/192 ($< 10^{-6}$)	0.8428 (−10)	51/61 (0.395)	0.8438 (+59)	65/6 (0.8713)
EnzBERT	$\leq 70\%$	95/69,666	0.8394 (0.8675)	0.8374 (−146)	96/242 ($< 10^{-6}$)	0.8391 (−26)	41/67 (0.0157)	0.8405 (+70)	76/6 (0.8682)
EnzBERT	$\leq 50\%$	95/69,666	0.8330 (0.8602)	0.8304 (−181)	132/313 ($< 10^{-6}$)	0.8324 (−38)	31/69 (1.8×10^{-4})	0.8343 (+89)	93/4 (0.8611)
EnzBERT	$\leq 30\%$	95/69,666	0.8256 (0.8515)	0.8232 (−167)	108/275 ($< 10^{-6}$)	0.8254 (−18)	39/57 (0.082)	0.8267 (+78)	80/2 (0.8524)

S3 Cross-ontology bridge and multi-kingdom population evaluations

Supplementary Table S7 summarizes the five constraint tables extracted from the OBO Foundry releases of the Gene Ontology, Uberon, the Plant Ontology, the Cell Ontology, and the NCBI Taxonomy, which the taxonomic and anatomical constraint modules load at run time. Supplementary Table S8 reports the 1,000-genome RefSeq bacterial audit, and Supplementary Table S9 the 88-proteome multi-kingdom evaluation.

The 1,000-genome audit (`paper-reproducibility/pipelines/P12_refseq1000/`) recomputes the structural metrics of six published GO predictors on 1,000 RefSeq bacterial assemblies from their per-genome prediction files with one evaluator; essential core-22, complex coherence, MetaCyc completeness, and `has_part` coherence reproduce the original audit's per-genome matrices for all 6,000 genome–tool pairs. The essential list holds 32 terms. The taxon columns count rows whose term is forbidden for the genome's lineage; an earlier version of this table counted every row whose term carries any taxon constraint, violated or not, which gives values 2.7 to 400 times larger. Experimental GOA annotations exist for 0.13 proteins per genome on average (985 genomes have none, after mapping RefSeq proteins to UniProtKB), which is too sparse for a genome-level CAFA evaluation, so no $F_{\max}$ or $S_{\min}$ is reported. GSPA runs Stage 0 and Stage 1 + 2A relaxation per genome with DIAMOND against Swiss-Prot without the nine benchmark genomes, Pfam via `hmmsearch`, and the InterProScan output of the original audit, with RefSeq operons and without gap claims; in the debug runs this took 4.5 to 10 minutes of wall time per genome on four cores, most of it `hmmsearch`. The GSPA array is still running, so the GSPA rows are marked as pending.

In the multi-kingdom evaluation, the proteomes of each domain were integrated in one shared-JVM batch under three configurations: Stage 0 noisy-OR without priors, and all Stage 1 priors with Stage 2A relaxation for one or for three iterations. Mean $F_{\max}$ increased over Stage 0 in every domain under both GSPA configurations. The paired bootstrap interval of the mean per-proteome difference excluded zero for Bacteria (47 proteomes) and Viruses (20) and included zero for Archaea (15) and Eukaryota (6). The bootstrap is computed by `paper-reproducibility/pipelines/P15_multikingdom/bootstrap_ci.py` from the per-proteome evaluation. The Stage 3 trigger rates in the last column come from re-running the three-iteration configuration of every proteome with the anomaly extractor (Supplementary Table S21).

Table S7. Summary of programmatic cross-ontology constraint tables in GSPA v2.1.0.

Resource File	Ontologies Bridged	Data Rows	Distinct Classes	Constraint Semantics & Examples
`go-taxon-constraints.tsv`	GO ↔ NCBITaxon	26,911	18,159 GO terms	13,819 `only_in_taxon` + 13,092 `never_in_taxon` rules (e.g., `GO:0009579` thylakoid)
`go-anatomical-constraints.tsv`	GO ↔ Uberon / PO / CL	2,715	2,674 GO; 1,294 Anat.	30 object properties (`results_in_development_of`, `occurs_in`, etc.)
`uberon-po-taxon-constraints.tsv`	Uberon / PO / CL ↔ NCBITaxon	8,331	1,247 Anat. classes	4,767 `only_in_taxon` + 3,564 `never_in_taxon` rules (e.g., `UBERON:0000948` heart → Metazoa)
`uberon-po-hierarchy.tsv`	Uberon / PO / CL DAG	7,722	2,855 Anat. classes	Transitive `is_a` and `part_of` edges across anatomical structures
`ncbi-taxon-hierarchy.tsv`	NCBI Taxonomy DAG	1,818	1,305 Taxon clades	Transitive taxonomic `is_a` lineage closure for benchmark clades

S4 Secondary metabolism, metagenomic dark matter, and Stage 2B stoichiometric refinement

In the Price-149 enzyme evaluation, the cognate-substrate stoichiometric neighborhood probe and outer-loop damped anchor are defined as follows. For each candidate pair (p, u), let $n_1(p, u)$ and $n_2(p, u)$ count chromosomal neighbors within ± 10 genes whose reactions share non-currency metabolites with u at 1 and 2 hops, $f_{\text{op}}(p)$ denote the co-directional operon fraction, and $s_{\text{nb}}(p, u) = \min(2.0, \sum_{|g'-g|\leq 10}(1 + 0.25|g' - g|)^{-1} P_{\text{cand}}(g', u))$ measure distance-weighted neighbor EC support. The local stoichiometric probe shift $\Delta L_{\text{probe}}(p, u) = 0.85 \min(1.5, 0.55n_1 + 0.20n_2) + 0.45 f_{\text{op}}(p)\mathbb{I}[n_1 + n_2 > 0] + 0.35\mathbb{I}[u \in \mathcal{R}_{\text{bal}}] - 0.50\mathbb{I}[u \notin \mathcal{R}_{\text{ec}}]$ and Stage 1 neighbor coherence shift $\Delta L_{\text{stage1}}(p, u) = 0.90 s_{\text{nb}}(p, u) + 0.50 f_{\text{op}}(p)\mathbb{I}[s_{\text{nb}}(p, u) > 0.20] + 0.30\mathbb{I}[u \in \mathcal{W}_{\text{kegg}}]$ are applied over $T = 3$ damped outer

Table S8. Structural audit of 1,000 **RefSeq bacterial genomes.** Rows for the six published predictors are recomputed from their per-genome prediction files with the score cut-offs of the original audit; every metric reproduces the audit's per-genome matrices. Essential completeness is the presence of the 22 core and 32 full essential GO terms in the propagated annotation (GO release 2025-03-16); complex coherence is the percentage of complex terms carried by at least two proteins; MetaCyc completeness is the percentage of annotated pathways with a complete EC combination; `has_part` coherence is 100 minus the percentage of missing `has_part`-required terms. Taxon violations count protein–term rows whose term is forbidden for the genome's lineage (propagated rows, leaf rows, and proteins with at least one violation). GSPA rows (Stage 0 and Stage 1 + 2A relaxation with DIAMOND, Pfam, and InterProScan claims, posterior ≥ 0.50) are pending while the array job runs. Experimental GO truth covers 0.13 proteins per genome on average, so no CAFA $F_{\max}$ or $S_{\min}$ is reported; the original audit logged no runtime or memory. Source: `results/pillar1_suite2_1000bacteria_sota.tsv` (pipeline P12).

Method	Genomes	Essential core-22 (%)	Essential full-32 (%)	Complex coherence (%)	MetaCyc completeness (%)	has_part coherence (%)	Taxon-violating rows, propagated	Taxon-violating rows, leaf	Proteins with a violation
DeepGO-SE / DeepGOMeta (2024)	1,000	95.34	83.09	79.83	11.59	88.55	109.4	78.0	66.0
SPROF-GO (2023)	1,000	99.91	83.67	79.18	0.18	89.28	505.3	204.0	174.6
DeepGraphGO (2021)	1,000	99.28	85.67	70.56	37.37	89.94	1,653.4	711.9	431.8
DeepFRI (2021)	1,000	100.00	81.25	98.87	9.43	84.52	39,836.9	19,708.8	2,093.1
TALE (2021)	1,000	87.24	75.19	65.73	19.63	86.83	1,381.4	703.2	566.8
InterProScan 5	1,000	97.36	86.65	61.89	40.35	90.28	2.3	1.5	1.5
GSPA Stage 0 (Noisy-OR)	*pending (array running; 4 of* 1,000 *genomes evaluated at aggregation)*								
GSPA Stage 1+2 (Full)	*pending (array running; 4 of* 1,000 *genomes evaluated at aggregation)*								

Table S9. Multi-kingdom evaluation on 88 **proteomes.** Proteins are scored against held-out Swiss-Prot GO annotations. Stage 0 claims come from DIAMOND against a reference that excludes the target species and genus and from Pfam (the 19 genomes of the 9- and 10-genome sets also carry InterProScan claims). $F_{\max}$ values are protein-centric and ancestor-propagated, averaged over the proteomes of each domain; the <30% column restricts evaluation to proteins whose best reference hit is below 30% identity (no per-protein identity file was available for the bacterial proteomes, so the bacterial value repeats the all-protein value). $\Delta F_{\max}$ is the mean per-proteome difference from Stage 0 with a 95% paired percentile bootstrap interval over proteomes (10,000 resamples, seed 20261019; `results/multikingdom_bootstrap_ci.tsv`, pipeline P15). The Stage 3 trigger rate is the pooled percentage of proteins referenced by at least one anomaly under the cap of 150 anomalies per proteome, computed from Stage 1 + 2A runs with three iterations (`results/pillar4_v2_trigger_rates.tsv`; Supplementary Table S21). Runtime and heap are per proteome in one shared-JVM batch. Source: `results/multikingdom_summary_by_domain.tsv` (`scripts/run_multikingdom_batch_unimatrix.py`).

Domain	Configuration	Proteomes / proteins	$F_{\max}$ (all)	$F_{\max}$ (<30%)	MF / BP / CC $F_{\max}$	Coverage	$\Delta F_{\max}$ [95% CI]	Runtime / heap	Stage 3 trigger
Archaea	Stage 0 noisy-OR	15/8,639	0.9125	0.5419	0.9460/0.9341/0.9146	0.9524	Reference	0.35 s / 4,746 MB	–
	GSPA Stage 1 + 2A (1 iteration)	15/8,639	0.9129	0.5422	0.9454/0.9345/0.9163	0.9524	+0.0004 [−0.0004, +0.0009]	2.29 s / 5,120 MB	–
	GSPA Stage 1 + 2A (3 iterations)	15/8,639	0.9128	0.5421	0.9452/0.9345/0.9163	0.9524	+0.0003 [−0.0009, +0.0011]	3.88 s / 5,253 MB	6.7%
Bacteria	Stage 0 noisy-OR	47/104,032	0.7987	0.7987	0.8326/0.8304/0.8322	0.8443	Reference	2.16 s / 4,479 MB	–
	GSPA Stage 1 + 2A (1 iteration)	47/104,032	0.8009	0.8009	0.8319/0.8316/0.8371	0.8443	+0.0021 [+0.0014, +0.0029]	12.66 s / 4,812 MB	–
	GSPA Stage 1 + 2A (3 iterations)	47/104,032	0.8004	0.8004	0.8313/0.8316/0.8371	0.8443	+0.0017 [+0.0007, +0.0026]	27.85 s / 4,898 MB	5.2%
Eukaryota	Stage 0 noisy-OR	6/6,155	0.7096	0.2459	0.8382/0.6709/0.7520	0.9079	Reference	1.13 s / 5,676 MB	–
	GSPA Stage 1 + 2A (1 iteration)	6/6,155	0.7117	0.2453	0.8380/0.6732/0.7555	0.9079	+0.0022 [−0.0001, +0.0052]	6.67 s / 4,232 MB	–
	GSPA Stage 1 + 2A (3 iterations)	6/6,155	0.7109	0.2453	0.8375/0.6727/0.7554	0.9079	+0.0014 [−0.0018, +0.0048]	10.13 s / 4,965 MB	6.5%
Viruses	Stage 0 noisy-OR	20/2,805	0.7222	0.3112	0.8303/0.7726/0.7004	0.7576	Reference	0.27 s / 4,841 MB	–
	GSPA Stage 1 + 2A (1 iteration)	20/2,805	0.7252	0.3116	0.8304/0.7747/0.7037	0.7576	+0.0030 [+0.0014, +0.0049]	0.42 s / 4,575 MB	–
	GSPA Stage 1 + 2A (3 iterations)	20/2,805	0.7252	0.3116	0.8304/0.7743/0.7037	0.7576	+0.0030 [+0.0014, +0.0049]	0.54 s / 4,462 MB	4.9%

iterations ($\lambda_t \in \{0.70, 0.42, 0.252\}$):

$$P^{(t)}_{\text{single}}(p,u) = \sigma_{\log}\Big(0.75\,\text{logit}\big(P^{(t-1)}(p,u)\big) + 0.25\,\text{logit}\big(P^{(0)}(p,u)\big) + \lambda_t\big(\Delta L_{\text{stage1}}(p,u) + 0.65\,\Delta L_{\text{probe}}(p,u)\big)\Big), \quad \text{(S1)}$$

and the multi-predictor posterior blends $0.55P^{(t)}_{\text{single}}(p,u) + 0.45P_{\text{NoisyOR}}(p,u)$. Supplementary Table S10 reports the biosynthetic gene cluster benchmark, Supplementary Table S11 the metagenome-assembled genome panel, and Supplementary Table S12 the 108-genome stoichiometric invariant audit, the comparison with six curated BiGG models, and the Price-149 arms split by metabolic stratum.

The biosynthetic gene cluster benchmark (`paper-reproducibility/pipelines/P13_bgc_antismash/`) uses the antiSMASH 8.0.4 output for the 108 genomes of the invariant audit. Every antiSMASH region is one cluster (1,292 regions in 106 genomes). Genes that antiSMASH marks `biosynthetic` are core loci (2,778), genes marked `biosynthetic-additional` are tailoring loci (4,683), the remaining genes inside regions are excluded, and the 480,236 genes outside every region are the negatives. A core gene takes the class of the antiSMASH rule that fired on it and a tailoring gene the classes of the protoclusters that span it, so a polyketide synthase in a hybrid region is scored against the polyketide term alone. Classes map to GO as follows: NRPS to GO:0019184 (nonribosomal peptide biosynthetic process), PKS to GO:0030639 (polyketide biosynthetic process), terpene to GO:0016114 (terpenoid biosynthetic process), siderophore (antiSMASH products NI-siderophore and NRP-metallophore) to GO:0019290 (siderophore biosynthetic process), RiPP to GO:0030152 (bacteriocin biosynthetic process), and all regions to GO:0044550 (secondary metabolite biosynthetic process) or any of the five class terms, because GO places GO:0019184 outside

GO:0044550. A gene receives a term when the highest posterior over the term and its `is_a` and `part_of` descendants is at least 0.50. Recall is pooled over genes, specificity is $1-\mathrm{FP}/N$ over the non-cluster genes, precision is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ with true positives over core and tailoring genes, and F_1 combines precision with pooled core-plus-tailoring recall; confidence intervals are percentile bootstraps over genomes (1,000 replicates). Stage 0 claims come from DIAMOND 2.1.16 (`blastp`, E-value $\leq 10^{-5}$, five targets) against the 83,592 Swiss-Prot entries with experimental GO annotations, HMMER 3.3.2 (`hmmsearch -cut_ga`) against Pfam-A with `pfam2go`, and InterProScan 5.61-93.0 with InterPro2GO (Pfam, TIGRFAM, HAMAP, PIRSF, SMART, CDD, PROSITE, PRINTS, and SFLD; Gene3D, SUPERFAMILY, PANTHER, and FunFam were omitted to save compute). Operons come from the region coordinates (same strand, gap ≤ 300 bp), and no gapseq gap table was supplied. GSPA v2.1.0 ran three configurations in batch mode under the asserted taxon Bacteria: Stage 0 alone, the operon prior alone, and all six priors with Stage 2A relaxation. The operon-only configuration reproduces the Stage 0 values to four decimals, and the full configuration moves 19 of the 1,385 core-gene class claims by more than 0.01, because the genomic context prior propagates only functions that a KEGG pathway requires and none of the five class terms occurs in the pathway table. Of the 2,568 class-term claims on core genes, 2,255 are DIAMOND alignments to distant experimentally annotated Swiss-Prot proteins whose identity-calibrated probability lies below 0.50 (median Stage 0 posterior 0.24); at a threshold of 0.30, core recall rises to 0.18 for PKS, 0.29 for terpenes, and 0.57 for siderophores, with Stage 0 and the full configuration still within 0.003 of each other (`results/pillar2_3_antismash_bgc_specificity_v2_thr030.tsv`; claim diagnostics in `paper-reproducibility/pipelines/P13_bgc_antismash/outputs/`).

The metagenome-assembled genome (MAG) panel (`paper-reproducibility/pipelines/P14_mag_pathway_gaps/`) uses single-sample SemiBin MAGs from Empty Quarter metagenomes. Of 3,835 MAGs labelled high quality, all have CheckM2 completeness $\geq 90\%$ and contamination $\leq 5\%$ and a GTDB-Tk r226 phylum, and 3,794 in 35 phyla also have a completed gapseq run and a Bakta annotation. From the 17 phyla with at least 10 eligible MAGs we drew a phylum-stratified sample of 100 MAGs (88 bacterial and 12 archaeal; four per phylum, the remaining 32 in proportion to phylum size; seed 20261019). Gap reactions are the SEED reactions of gapseq 1.4.0's gap-filled model that are absent from its draft model, mapped onto Bakta loci together with gapseq's own GPR genes and sub-threshold tblastn hits. Stage 0 claims come from DIAMOND 2.1.16 (`blastp`, E-value $\leq 10^{-5}$, five targets) against 83,592 Swiss-Prot entries with experimental GO annotations and from HMMER 3.3.2 (`hmmsearch -cut_ga`) against Pfam-A with `pfam2go`; InterProScan was not run. GSPA v2.1.0 ran Stage 0 (`-enable-priors none`) and all six priors with Stage 2A relaxation, the gap table, Bakta operons (same strand, gap ≤ 300 bp), and RLGC (`-rlc-suggester` with gapseq's reaction graph, $k = 2$, bandwidth 5 kb), under the asserted taxon of Bacteria or Archaea and the matching essential profile. Across the 100 MAGs, 1,258 new fills reached posterior 0.50 (12.6 per MAG), mean MetaCyc pathway completeness rose from 0.140 to 0.274 (gapseq's own value is 0.945), and essential completeness rose from 0.782 to 0.932 (Supplementary Table S11). Of the 1,258 fills, 742 fall on the locus of gapseq's own sub-threshold hit for the reaction, 491 on a locus that gapseq had not linked to the reaction, and 25 on reactions without any gapseq hit; no fill concerns a reaction to which gapseq attached a gene. The gap-filling prior produced 1,254 of the fills at $q \geq 0.30$ and RLGC 18 (4 at $q \geq 0.50$).

The invariant audit compares four networks built from the same DeepProZyme draft (reactions with any mapped EC scored at least 0.50): the draft without repair, METEOR v8's archived negative-cost MILP solutions, the Grow-Support positive-cost threshold repair re-derived from the archived code, and Stage 2B. Each network is audited on two media. METEOR's own medium uses the FVA-tight bounds of its MILP with every non-medium exchange open for uptake and secretion closed; the audit medium is the Stage 2B minimal medium with the 36-compound ModelSEED supplement, with uptake restricted to those compounds and secretion open. No METEOR or Grow-Support network grows on the strict audit medium, so flux certification on that medium adds the smallest set of further uptakes needed for growth, while the feasibility column reports the strict medium. Under both media, 89% of METEOR's added reactions carry no flux in a parsimonious FBA solution at the minimum biomass flux, compared with 1.6% (own medium) and 13.8% (audit medium) of the Grow-Support additions, and all of Stage 2B's additions carry flux. GSPA's draft core is seeded from METEOR's archived active set, so the GSPA row compares repair policies on a shared draft core (`paper-reproducibility/pipelines/P07_invariant_audit/README.md`).

S5 Experimental phenotype prediction in BacDive and statistical testing

The BacDive benchmark uses the two comparison tables published with gapseq: 2,332 ProTraits carbon-source assays from 791 strains and 50 substrates, and 17,496 BacDive enzyme assays from 3,164 strains after keeping assays with a definitive activity and calls from all three reconstructions. In the five-fold genus-blocked cross-validation, each genus (the first word of the organism name) is assigned to fold MD5(genus) mod 5, so that no genus contributes

Table S10. Biosynthetic gene cluster class recovery and non-cluster specificity on the 1,292 **antiSMASH regions of the** 108 **invariant-audit genomes.** Every antiSMASH 8.0.4 region is one cluster; core loci are the genes antiSMASH marks `biosynthetic` and tailoring loci the genes it marks `biosynthetic-additional`, each scored against the GO term of its own class (NRPS, GO:0019184; PKS, GO:0030639; terpene, GO:0016114; siderophore, GO:0019290; RiPP, GO:0030152; all regions, GO:0044550 or any class term), and the 480,236 genes outside every region are the negatives. A gene receives a term when the highest posterior over the term and its descendants is at least 0.50. Recall is pooled over genes (95% percentile bootstrap over genomes in parentheses), specificity is $1 - \mathrm{FP}/N$ on the non-cluster genes, and F_1 combines precision with pooled core-plus-tailoring recall. Stage 0 is the Noisy-OR of DIAMOND, Pfam, and InterProScan claims; GSPA is all six Stage 1 priors with Stage 2A relaxation under the asserted taxon Bacteria. The operon-only configuration reproduces the Stage 0 values to four decimals and is omitted. The last column is the fraction of core genes with any claim for the class term before thresholding. Source: `results/pillar2_3_antismash_bgc_specificity_v2.tsv` (pipeline P13, `benchmark/unified/pillar2_fba_lro_price149/bgc_benchmark.py`).

Class	Genomes / clusters	Core / tailoring loci	Core recall, Stage 0 (95% CI)	Core recall, GSPA	Tailoring recall, Stage 0 / GSPA	Specificity, GSPA	F_1, Stage 0 / GSPA	Core genes with claim
NRPS	72/359	778/1,877	0.069 (0.045–0.104)	0.069	0.019/0.019	0.9996	0.061/0.061	0.71
PKS	75/270	649/1,645	0.003 (0.000–0.008)	0.003	0.002/0.002	0.9999	0.004/0.004	0.54
Terpene	106/303	426/649	0.157 (0.123–0.192)	0.157	0.133/0.133	0.9993	0.196/0.196	0.58
Siderophore	63/117	305/462	0.528 (0.448–0.627)	0.528	0.078/0.078	0.9998	0.377/0.377	0.76
RiPP	93/314	570/554	0.000 (0.000–0.000)	0.000	0.000/0.000	1.0000	0.000/0.000	0.00
All regions	106/1,292	2,778/4,683	0.095 (0.079–0.113)	0.093	0.028/0.028	0.9986	0.093/0.092	0.50

Table S11. Pathway gap resolution in 100 **high-quality metagenome-assembled genomes from the Empty Quarter.** A fill is a pair of a gapseq gap reaction (in the gap-filled model and absent from the draft) and a Bakta locus that carries one of the reaction's functions after Stage 1 + 2A with gaps, operons, and RLGC (posterior or RLGC q-value ≥ 0.30 or ≥ 0.50), counted only when the locus had Stage 0 posterior below 0.30 for every function of the reaction; the ≥ 0.50 fills are a subset of the ≥ 0.30 fills. MetaCyc pathway completeness is the mean, over gapseq-predicted MetaCyc pathways, of the fraction of EC-bearing reactions with a locus at posterior ≥ 0.50 (Stage 0 $\rightarrow$ Stage 1 + 2A with RLGC); essential completeness is the fraction of the domain's essential GO terms covered at posterior ≥ 0.50. Means are over MAGs. Source: `results/pillar3_dark_matter_mags_v2.tsv` (pipeline P14, `benchmark/unified/pillar3/mag_pathway_gaps.py`).

Domain	Phylum	MAGs	Fills (≥ 0.30)	Fills (≥ 0.50)	Fills per MAG (≥ 0.30 / ≥ 0.50)	MetaCyc completeness	Essential completeness
Bacteria	Actinomycetota	17	231	230	13.6/13.5	0.160 → 0.312	0.774 → 0.925
Bacteria	Pseudomonadota	12	94	91	7.8/7.6	0.213 → 0.398	0.800 → 0.948
Bacteria	Bacillota	6	40	40	6.7/6.7	0.217 → 0.378	0.870 → 0.974
Bacteria	Bacteroidota	6	94	94	15.7/15.7	0.144 → 0.281	0.766 → 0.958
Bacteria	Acidobacteriota	5	77	76	15.4/15.2	0.141 → 0.277	0.806 → 0.944
Bacteria	Bacteroidota_A	5	69	69	13.8/13.8	0.125 → 0.266	0.806 → 0.938
Bacteria	Chloroflexota	5	70	70	14.0/14.0	0.107 → 0.228	0.788 → 0.900
Bacteria	Gemmatimonadota	5	92	88	18.4/17.6	0.095 → 0.226	0.744 → 0.938
Bacteria	Myxococcota	5	91	89	18.2/17.8	0.136 → 0.301	0.788 → 0.963
Bacteria	Patescibacteriota	5	53	53	10.6/10.6	0.071 → 0.087	0.588 → 0.844
Bacteria	Planctomycetota	5	100	100	20.0/20.0	0.114 → 0.234	0.744 → 0.925
Bacteria	Bdellovibrionota	4	53	52	13.3/13.0	0.142 → 0.286	0.773 → 0.938
Bacteria	Deinococcota	4	47	45	11.8/11.3	0.161 → 0.342	0.797 → 0.891
Archaea	Halobacteriota	4	26	26	6.5/6.5	0.079 → 0.129	0.827 → 0.933
Archaea	Thermoplasmatota	4	35	35	8.8/8.8	0.054 → 0.135	0.846 → 0.923
Archaea	Thermoproteota	4	45	45	11.3/11.3	0.076 → 0.118	0.808 → 0.942
Bacteria	Verrucomicrobiota	4	55	55	13.8/13.8	0.139 → 0.306	0.781 → 0.938
	All 17 phyla	100	1,272	1,258	12.7/12.6	0.140 → 0.274	0.782 → 0.932

to both training and test data. Within each training split, the conditional probabilities of a positive phenotype given the gapseq call ($p_{g,\mathrm{cal}}$) and the ModelSEED call ($p_{m,\mathrm{cal}}$) and the assay prevalence $a_{\mathrm{prev}}(y)$ are estimated with additive smoothing, and they are combined with the Track 1 posterior and the coupling score $\psi(s, y)$ into a fold-local pooled probability

$$\operatorname{logit} p_{\mathrm{pool}}(s, y) = \beta_g \operatorname{logit} p_{g,\mathrm{cal}} + \beta_m \operatorname{logit} p_{m,\mathrm{cal}} + \beta_a \operatorname{logit} a_{\mathrm{prev}}(y) + \beta_1 \operatorname{logit} P_{\mathrm{Track1}}(s, y) + \beta_\psi \, \psi(s, y), \tag{S2}$$

with $(\beta_g, \beta_m, \beta_a, \beta_1, \beta_\psi) = (0.45, 0.25, 0.15, 0.15, 0.50)$ for carbon sources and $(0.48, 0.24, 0.14, 0.14, 0.45)$ for enzymes. The blended reconstruction call used for ψ is $0.50g + 0.32m + 0.18c$ for carbon sources and $0.52g + 0.30m + 0.18c$ for enzymes. The GSPA hybrid trains a gradient-boosted classifier on the three calls, these fold-local quantities, the strain's call profile, and a one-hot assay indicator, and reports 0.88 times its output plus $0.12\, p_{\mathrm{pool}}$. The supervised baselines are re-implementations in scikit-learn, after the designs of Traitar (L_1-regularized linear SVM with probability calibration), BacDive-AI (random forest), and DiASPora (gradient-boosted trees), trained on the three binary calls and a one-hot assay indicator. In Track 2, 30% of each strain's assays (selected by an MD5 hash of strain, assay, and row index) are revealed and the remaining 70% are evaluated. All Track 2 quantities, including p_{pool}, are estimated

Table S12. Stoichiometric invariant audit, curated reference models, and Price-149 strata. (a) All four networks start from the DeepProZyme draft at threshold 0.50 for each of 108 genomes. Added reactions are internal reactions absent from the draft; an added reaction is flux-certified when it carries flux ($|v| > 10^{-6}$) in a parsimonious FBA solution that reaches $v_{\text{bio}} \geq 0.1$ within the network's own reactions. "Own" is METEOR's medium (FVA-tight bounds with every non-medium exchange open for uptake); "audit" is the Stage 2B minimal medium with the 36-compound ModelSEED supplement, to which the greedy-minimal set of non-medium uptakes needed for growth is added for flux certification because no METEOR or Grow-Support network grows on the strict medium. "Grows, audit medium" is measured on the strict audit medium; for the raw draft it is the archived growth flag. METEOR repairs are archived solutions, not re-runs. GSPA's draft core is seeded from METEOR's archived active set, so the GSPA row compares repair policies on a shared draft core. [a]Mixed column: MEMOTE mass-imbalance counts over the active network for the first three rows and over added reactions for GSPA. (b) Means over six organisms with curated BiGG models (iML1515, STM_v1_0, iYL1228, iJN1463, iYS854, iYO844). (c) Price-149 arms as in Table 6 of the main text, split by the metabolic role of the target reaction. Sources: `results/pillar2_invariant_audit_summary.tsv` (pipeline P07), `results/pillar2_curated6_gems_real.tsv`, `results/pillar2_price149_by_stratum.tsv` (pipeline P08).

(a) Stoichiometric invariant audit over 108 *genomes (means per genome)*

Method	Active	Flux-carrying / zero-flux (own)	Flux-carrying (audit)	Added	Certified / padding (own, % padding)	Certified / padding (audit, % padding)	Imbalanced added	Imbalanced added or active[a]	Dead-end active	Grows, audit medium (%)	Grows, own medium (%)
DeepProZyme draft (≥ 0.50), no repair	2,809.5	0.0 / 2,809.5	–	0.00	0.0 / 0.0 (0.0)	0.0 / 0.0 (0.0)	0.00	2,621.7	9.75	0.0	0.0
METEOR v8 (negative-cost MILP, archived)	3,121.0	312.0 / 2,809.0	310.8	611.96	65.2 / 546.8 (89.4)	65.2 / 546.8 (89.4)	6.25	2,597.8	2.36	0.0	100.0
Grow-Support repair (positive-cost LP)	2,584.8	316.0 / 2,268.9	302.3	52.71	51.9 / 0.8 (1.6)	45.4 / 7.3 (13.8)	6.25	2,278.9	2.23	0.0	100.0
GSPA Stage 2B (positive-cost LP + pFBA)	2,815.2	300.6 / 2,514.6	300.6	5.70	5.7 / 0.0 (0.0)	5.7 / 0.0 (0.0)	0.00	0.0	0.00	100.0	–

(b) Six curated BiGG genome-scale models (mean over the six organisms)

Reconstruction	Models with growth	EC precision / recall / F_1	BiGG reaction precision / recall / F_1
CarveMe	6/6	0.8217 / 0.8927 / 0.8536	0.6693 / 0.7305 / 0.6955
Reconstructor (ModelSEED namespace)	4/6	0.4387 / 0.8787 / 0.5844	– (ModelSEED identifiers)
DeepProZyme + METEOR v8	6/6	0.4383 / 0.8842 / 0.5851	0.1655 / 0.2900 / 0.2091

(c) Price-149 by metabolic stratum, CLEAN as the row predictor: correct Top-1 ECs (dataset-level macro $F_{\max}$*)*

Stratum	Proteins	Stand-alone	+ METEOR v8	+ SLR + RLGC	+ GSPA single-predictor	+ Stage 0 pooling	Pool without this predictor	+ GSPA multi-predictor
A: core biomass	91	46 (0.4740)	47 (0.4775)	46 (0.4884)	47 (0.5095)	49 (0.5308)	46 (0.5467)	54 (0.5318)
B: catabolic, assay-specific	55	13 (0.3103)	13 (0.3009)	13 (0.3103)	15 (0.3062)	14 (0.3470)	7 (0.3729)	15 (0.3522)
C: secondary or peripheral	2	1 (0.5000)	1 (0.5000)	1 (0.5000)	1 (0.5000)	1 (0.5000)	1 (0.5000)	1 (0.5000)

on the revealed assays only. For carbon sources, $\operatorname{logit} P_{\text{Track2}} = 0.72 \operatorname{logit} p_{\text{pool}} + 0.28 \operatorname{logit} \hat{a}(y) + 0.85\, \psi_{\text{rev}}(s, y)$, where $\hat{a}(y)$ is the smoothed positive rate of assay y among revealed assays and ψ_{rev} applies the coupling matrix Ψ to the strain's revealed outcomes coded as ± 1. For enzymes, $\operatorname{logit} P_{\text{Track2}} = 0.70 \operatorname{logit} p_{\text{pool}} + 0.30 \operatorname{logit} \hat{g}(s, y)$, where $\hat{g}$ is the smoothed revealed positive rate of the enzyme in the strain's genus, or of the enzyme overall when the genus has no revealed assay of it. The decision threshold is tuned on the revealed assays, and the Track 2 hybrid reports 0.65 times a gradient-boosted classifier trained on the revealed assays plus $0.35\, P_{\text{Track2}}$. Supplementary Table S13 reports per-fold F_1 for both assay panels; the code is `benchmark/unified/pillar3_bacdive/run_bacdive_tracks1_2.py`.

Paired comparisons of binary outcomes use the exact two-sided McNemar test, computed as a binomial test of the discordant counts b and c against $\text{Bin}(b + c, 0.5)$; this applies to the BacDive, Price-149, 95-genome, leave-subunit-out, and Stage 3 comparisons. Confidence intervals are percentile bootstrap intervals: over proteomes for the multi-kingdom $F_{\max}$ differences (10,000 resamples, seed 20261019) and over proteins for Price-149 Top-1 accuracy (2,000 resamples, seed 42).

Table S13. Five-fold genus-blocked cross-validation in BacDive. Each genus is assigned to one fold by its MD5 hash. Columns give F_1 on each held-out fold, the mean ± sample standard deviation over folds, and the pooled out-of-fold F_1 with balanced accuracy and ROC-AUC in parentheses. The supervised classifiers are re-implementations (an L_1-regularized linear SVM, a random forest, and a gradient-boosted decision tree) over the three binary reconstruction calls of gapseq, ModelSEED, and CarveMe and a one-hot assay indicator, after the designs of Traitar, BacDive-AI, and DiASPora; the GSPA hybrid trains the same GBDT on GSPA posteriors. gapseq and GSPA Track 1 use no phenotype labels, and their per-fold values evaluate the same predictions on each fold. Strain counts are distinct strains in the gapseq comparison tables after the assay filter. Sources: `results/pillar3_bacdive_per_fold_cv.tsv`, `results/pillar3_bacdive_track1_summary.tsv`.

Assay panel	Method	Fold 1 F_1	Fold 2 F_1	Fold 3 F_1	Fold 4 F_1	Fold 5 F_1	Mean ± SD	Pooled F_1 (BAcc / ROC-AUC)
Carbon-source growth 791 strains 2,332 assays	gapseq calls (unsupervised reference)	0.8543	0.7731	0.7991	0.5682	0.8280	0.7645 ± 0.1139	0.7952 (0.8082/0.8082)
	GSPA Track 1 (no labels)	0.8543	0.7712	0.8331	0.5747	0.8293	0.7725 ± 0.1148	0.8094 (0.8239/0.8620)
	L1-SVM on reconstruction calls (after Traitar)	0.8480	0.8765	0.8727	0.7708	0.8653	0.8467 ± 0.0438	0.8571 (0.8638/0.9185)
	Random forest on reconstruction calls (after BacDive-AI)	0.8709	0.8720	0.8397	0.7174	0.8646	0.8329 ± 0.0659	0.8448 (0.8530/0.9222)
	GBDT on reconstruction calls (after DiASPora)	0.8582	0.8819	0.8582	0.7817	0.8238	0.8408 ± 0.0390	0.8483 (0.8564/0.9268)
	GSPA posteriors + GBDT	0.8897	0.9040	0.8919	0.7905	0.8617	0.8676 ± 0.0458	0.8778 (0.8848/0.9567)
Diagnostic enzymes 3,164 strains 17,496 assays	gapseq calls (unsupervised reference)	0.7702	0.7629	0.7747	0.7939	0.7725	0.7748 ± 0.0115	0.7753 (0.7461/0.7461)
	GSPA Track 1 (no labels)	0.7735	0.7644	0.7751	0.7918	0.7722	0.7754 ± 0.0100	0.7757 (0.7483/0.7465)
	L1-SVM on reconstruction calls (after Traitar)	0.8620	0.8561	0.8562	0.8714	0.8517	0.8595 ± 0.0076	0.8593 (0.8014/0.8715)
	Random forest on reconstruction calls (after BacDive-AI)	0.8477	0.8371	0.8490	0.8553	0.8387	0.8456 ± 0.0076	0.8458 (0.7911/0.8593)
	GBDT on reconstruction calls (after DiASPora)	0.8465	0.8579	0.8560	0.8655	0.8493	0.8550 ± 0.0075	0.8550 (0.8108/0.8699)
	GSPA posteriors + GBDT	0.8670	0.8658	0.8708	0.8820	0.8696	0.8710 ± 0.0064	0.8712 (0.8256/0.9098)

S6 Stage 3: tool schemas, benchmark construction, model settings, and seed stability

Stage 3 gives the language model five deterministic, read-only tools, defined in `benchmark/agent/run_gspa_agent.py`:

1. `inspect_operon_neighborhood(protein_id)` returns the operon membership of a candidate protein and the annotated functions of its neighbouring genes.
2. `inspect_pathway_or_complex_gap(anomaly_id)` returns the context of the missing reaction or subunit and the candidate proteins of the anomaly.
3. `query_protein_evidence(protein_id)` returns the sequence, domain, structure, language-model, and genomic-context claims of a candidate protein.
4. `check_symbolic_constraints(go_term,organism_taxon)` checks a GO term against the taxonomic (SAT) and Uberon/PO/CL anatomical constraints for the organism and lists the nearest valid ancestors when the term is forbidden.
5. `propose_resolution(anomaly_id,action,protein_id,function_id,confidence,rationale)` submits the decision, one of PROMOTE, RELAX, DEMOTE, or ABSTAIN, with a confidence in $[0.50, 0.95]$ for PROMOTE and RELAX or in $[0.05, 0.25]$ for DEMOTE. The symbolic guardrail verifies the proposed term before the decision is accepted.

The benchmark (version 2) is built from the anomaly records that the integrator itself writes (`paper-reproducibility/pipelines/P10_stage3_agent/`). Seven genomes were integrated with all priors, Stage 2A relaxation, operons, and gapseq gaps: *E. coli* K-12, *B. subtilis*, *P. aeruginosa*, and *M. tuberculosis* with DIAMOND, Pfam, and InterProScan

claims, and *S. cerevisiae*, *C. elegans*, and *A. thaliana* with DIAMOND and Pfam claims. For pathway gaps, every annotated pathway step covered by a single protein was held out in turn by deleting that protein's claims for the step's term and its descendants; the step was added to the gap table, the integrator was re-run with the anomaly cap lifted, and the extractor's own pathway-gap record was taken. Of 597 hold-outs, 50 produced a record in which the held-out protein re-entered the candidate list through the dark-matter suggester. These were split by SHA-256 order into a true-positive half and a decoy half before any decoy was run; in a decoy the held-out protein is removed from the genome and the operon table. Decoys were matched one-to-one to true positives on candidate count and maximum candidate posterior (± 0.05). For ontology conflicts, 40 proteins per genome received two claims for a GO term that the guardrail forbids for the organism, at raw scores on a grid from 0.40 to 0.85, in Stage 1 runs without Stage 2A (with Stage 2A, all 280 injected violations were relaxed or removed before extraction). Genuine conflicts are the extractor's borderline-conflict records for injected terms (posterior in $[0.35, 0.50)$); traps are borderline-conflict records from the same runs whose term is supported by the truth annotations and valid for the organism, matched on candidate count, supporting-claim count, and posterior (± 0.05). The benchmark holds 20 true positives and 20 decoys from four bacteria and 20 genuine conflicts and 20 traps. Incomplete-complex and borderline-essential anomalies could not be constructed in this way: of 7 complex and 32 essential hold-outs, at most one produced a record containing the held-out protein, in line with their rarity in natural data (6 and 83 of 31,839 uncapped anomalies over the 88 multi-kingdom proteomes). GO labels shown to the model are the `name` fields of GO release 2026-03-25, and 8 records whose target term is absent from the guardrail's GO release (2025-06-01) were excluded.

Every protein identifier the model can see (candidates, operon neighbours, anchors) is replaced by an opaque token `P####`, operons by `O####`, and anomaly identifiers by `A###`, assigned in SHA-256 order, and candidate order is shuffled by SHA-256 within each anomaly; no gene symbol or accession appears in the model input. The mapping is in `results/pillar4_v2_id_mapping.tsv`, and the leak checks in `results/pillar4_v2_leak_checks.tsv` show identical model-visible fields for positives and negatives in both categories and anomaly-level AUCs of 0.47 to 0.53 for true positives against decoys on the maximum posterior, the maximum suggestion score, and the fraction of candidates in the pathway operon. In the conflict category, every genuine item carries the integrator's consistency prior (-0.68) and no trap does, so this category tests whether a method uses that symbolic signal.

The model is `Qwen/Qwen3.8-27B-FP8`, served by vLLM on two NVIDIA RTX 4090 GPUs with a context of 131,072 tokens, reasoning mode off, and at most 450 output tokens per turn. The configurations are zero-shot (one turn, only `propose_resolution`, no guardrail), fast (two turns, guardrail), guarded (six turns, guardrail; seed 42 at temperature 0.0 and seeds 43 and 44 at temperature 0.25), and unguarded (six turns, no guardrail, seed 42). Full transcripts, including every request and response with timestamps and the SHA-256 of the agent code in the header, are in `results/pillar4_v2_transcripts_<configuration>_<seed>.jsonl`, and those of the LRO-50 runs in `results/pillar4_v2_transcripts_lro50_<configuration>_<seed>.jsonl`. Supplementary Table S14 reports the three seeds of the guarded agent and the per-category decisions of every configuration, Supplementary Table S20 the paired tests, and Supplementary Table S21 the trigger rates. Supplementary Table S15 reports the isolated-protein CAFA evaluation (`results/suite1_cafa6_prospective_summary.tsv`).

S7 Pan-phylum AllTheBacteria analysis

The AllTheBacteria analysis (`paper-reproducibility/pipelines/P11_allthebacteria/`) runs in four stages on KAUST Ibex. Stage A calls ORFs with pyrodigal, records co-directional neighbour chains, removes exact duplicates, and clusters each batch with MMseqs2 `easy-linclust` (90% identity, 80% coverage); every ORF inherits the scores of its cluster representative. Stage B aligns every representative with DIAMOND (`blastp -sensitive -k 25`, E-value $\leq 10^{-3}$) against the experimentally annotated Swiss-Prot training set of the CAFA (Kaggle) release, which gives the maximum identity of each representative and the DIAMOND k-NN GO transfer, and scores the representative with the GSPA-Seq channels. Stage C writes per-genome claims and KEGG-coverage gap claims (pathways with 40–100% of their GO terms covered at ≥ 0.45; these are not gapseq gaps) and runs the integrator. Stage D runs the same integrator call on every genome of each completed batch, 96 genomes per JVM with a shared ontology load; a test on 20 genomes reproduced all 40 Stage C outputs and all 40 regenerated inputs byte for byte.

Supplementary Table S16 reports the alignment survey over 72,533 genomes, 11,632 species, and 37 phyla. Across 292 million ORFs, 41.0% had less than 30% identity to any training protein (85.8% of these had no alignment) and a further 27.0% had 30–50% identity. The dark fraction ranged from 36.5% in Actinomycetota and 40.4% in Pseudomonadota to 60.8–66.6% in Spirochaetota, Cyanobacteriota, Bacteroidota, and Planctomycetota. These counts depend only on the Stage B alignments.

Supplementary Table S17 reports the integrator results. The integrator call is `gspa-unified.jar` v2.1.0 `integrate` with all six priors, three iterations, and Stage 2A relaxation (`-enforce-stage2 -enforce-consistency-mode`

Table S14. Stage 3 on benchmark v2: seed stability and per-category decisions. The benchmark holds 80 anomalies extracted by the integrator from seven genomes: 20 pathway-gap true positives (the held-out protein re-entered the candidate list), 20 matched decoys (the held-out protein removed from the genome), 20 genuine borderline ontology conflicts, and 20 valid borderline terms (traps). (a) Precision, recall, and F_1 refer to promotions of the held-out protein; accuracy is over all 80 items; violations are accepted decisions whose term fails the guardrail. The mean row gives the mean ± population standard deviation over the three seeds. LRO-50 is the unchanged set of 50 ambiguous leave-reaction-out gaps with de-identified prompts. (b) A decoy is correct when the agent abstains or demotes; a genuine conflict is correct when the term is demoted or relaxed to a guardrail-valid ancestor (the lenient count also accepts abstention); a trap is correct when the term is kept. Model: `Qwen/Qwen3.8-27B-FP8` served by vLLM on two NVIDIA RTX 4090 GPUs. Sources: `results/pillar4_v2_multiseed_summary.tsv`, `results/pillar4_v2_per_category.tsv` (pipeline P10).

(a) Guarded agent across three seeds: 80-anomaly benchmark and LRO-50

Seed	Temperature	Precision	Recall	F_1	Accuracy	Violations (%)	LRO-50 Hit@1 (%)	LRO-50 MRR
42	0.00	0.7059	0.6000	0.6486	0.7750	0.0	84.0	0.8923
43	0.25	0.6471	0.5500	0.5946	0.7125	0.0	86.0	0.8990
44	0.25	0.5000	0.5000	0.5000	0.6625	0.0	88.0	0.9130
Mean ± SD	–	0.6177 ± 0.0866	0.5500 ± 0.0408	0.5811 ± 0.0614	0.7167 ± 0.0460	0.0 ± 0.0	86.0 ± 1.6	0.9014 ± 0.0086

(b) Correct decisions per category on the 80-anomaly benchmark (lenient count in parentheses)

Configuration	Pathway gap, true positive	Pathway gap, decoy	Conflict, genuine	Conflict, trap
Stage 1 + 2 only (abstain on every anomaly)	0/20	20/20	0/20 (20)	20/20
Posterior ≥ 0.35 + guardrail	0/20	0/20	20/20	20/20
Zero-shot, 1 turn, no tools, no guardrail (seed 42)	0/20	2/20	0/20 (20)	20/20
Fast mode, 2 turns, guardrail (seed 42)	3/20	20/20	11/20 (20)	16/20
Guarded agent, 6 turns (seed 42, $T = 0.0$)	12/20	16/20	20/20	14/20
Guarded agent, 6 turns (seed 43, $T = 0.25$)	11/20	15/20	19/20 (20)	12/20
Guarded agent, 6 turns (seed 44, $T = 0.25$)	10/20	12/20	19/20 (20)	12/20
Agent without guardrail, 6 turns (seed 42)	12/20	13/20	17/20 (19)	6/20

Table S15. Complete 17-method prospective CAFA time-based evaluation across 272 post-t_0 targets (`suite1_cafa6_prospective_summary.tsv`).

Method	**Targets** (n)	**MF** $F_{\max}$ (f_w)	**BP** $F_{\max}$ (f_w)	**CC** $F_{\max}$ (f_w)	**Mean** $F_{\max}$	**Mean** f_w	**Violations** (V)
Naive Term-Frequency Prior (CAFA Baseline 1)	272	0.4156 (0.4074)	0.2305 (0.1865)	0.4430 (0.2813)	0.3631	0.2917	43
InterPro Domain Logistic Regression (InterPro-LR)	272	0.4184 (0.4036)	0.2325 (0.2246)	0.1378 (0.1174)	0.2629	0.2485	76
DeepFRI Graph Convolutional Network	272	0.2790 (0.2729)	0.2568 (0.2247)	0.4666 (0.3669)	0.3341	0.2882	12
eggNOG-mapper v2 Orthology	272	0.4328 (0.4320)	0.2285 (0.2411)	0.4082 (0.3799)	0.3565	0.3510	22
DeepGOCNN (1D-CNN Sequence Model, Retrained)	272	0.4453 (0.4340)	0.2678 (0.2132)	0.5950 (0.4655)	0.4360	0.3709	19
ProteInfer Dilated Residual CNN	272	0.4406 (0.4404)	0.3161 (0.3267)	0.4750 (0.4530)	0.4106	0.4067	2
FoldSeek Structure Alignment (3Di+AA)	272	0.5693 (0.5760)	0.3246 (0.3029)	0.5492 (0.4091)	0.4810	0.4293	48
ESM-2 3B Fine-Tuned PLM	272	0.5714 (0.5526)	0.3641 (0.3310)	0.6609 (0.5426)	0.5321	0.4754	12
ProstT5 Structure-Aware PLM	272	0.5890 (0.5847)	0.3287 (0.2839)	0.6711 (0.5666)	0.5296	0.4784	86
ESM-2 Embedding k-NN (goPredSim, Retrained)	272	0.6136 (0.6144)	0.3499 (0.3048)	0.6417 (0.5194)	0.5351	0.4795	14
DIAMOND BLAST-kNN (CAFA Baseline 2)	272	0.6099 (0.6186)	0.3731 (0.3502)	0.6048 (0.4952)	0.5292	0.4880	27
ESM-2 MLP (Flat BCE, Retrained)	272	0.6441 (0.6320)	0.3192 (0.2894)	0.7045 (0.5928)	0.5559	0.5048	26
DeepGOPlus (1D-CNN + DIAMOND-kNN, Retrained)	272	0.6238 (0.6271)	0.3869 (0.3679)	0.6522 (0.5327)	0.5543	0.5092	22
ESM-2 C-HMCNN (Hierarchical MCLoss, Retrained)	272	0.6502 (0.6378)	0.3360 (0.3037)	0.6961 (0.5886)	0.5608	0.5100	22
GSPA-Seq (Val-Trained Per-Protein Integrator)	272	0.6621 (0.6628)	0.3964 (0.3665)	0.6933 (0.5942)	0.5839	0.5412	644
GSPA Stage 0 (Calibrated Noisy-OR)	272	0.6764 (0.6772)	0.3954 (0.3657)	0.6944 (0.5955)	0.5887	0.5461	160
GSPA Full (Stages 1 + 2A Relaxation)	272	0.6647 (0.6573)	0.3929 (0.3676)	0.6903 (0.5890)	0.5826	0.5380	0

`relax -lite`), compared with Stage 0 noisy-OR on the same claims. The Stage D snapshot holds all 7,009 genomes of the 15 wave-1 batches finished on 2026-10-19 (29 phyla, 3,965 species); run (a) holds the 4,592 genomes that Stage C had integrated, the first 20 genomes by sample identifier in each of 230 batches. In the snapshot, 15,603 proteins (0.06% of ORFs) gained a first annotation above 0.50, mostly through the essentiality prior and in the twilight regime; no zero-alignment protein was annotated, and the gap-filling prior fired on no row. Of the 875,786 suppressed pairs, 284,481 were taxon-forbidden and 521,301 were genome-wide co-annotation conflicts.

Several properties of the batch pipeline bound these numbers. No ESM-2 3B forward pass was run for any AllTheBacteria representative: representatives that occur in the training or benchmark sets use cached embeddings, and all others receive a bitscore-weighted interpolation of the embeddings of their DIAMOND hits, so zero-alignment ORFs have no language-model channel. No `hmmsearch` was run; the InterPro and Pfam channel copies the domain lists of Swiss-Prot hits with at least 35% identity and is also empty for zero-alignment ORFs, which therefore rely on the DeepGOCNN channel alone. Stage C writes each claim with a single merged source and one supporting claim, so correlation-grouped pooling has one group per pair, and the rescue paths that require two supporting claims or a candidate in the gap-claim window rarely apply. Taxon-forbidden suppression requires the genome's taxid in the integrator's bundled NCBI hierarchy, which holds 1,302 taxa; only 43 of the 6,460 distinct taxids of the collection are in it, and the other genomes are constrained only through genome-wide co-annotation conflicts. The `priors_fired` field holds each prior's boost at the final refinement state, so attributions to priors that fired only in earlier, damped iterations are missed.

Stage D over all completed batches (43 wave-1 batches with 20,683 genomes and 204 wave-2 batches with 449,653 genomes, about 33,700 CPU hours at 258 CPU seconds per genome) is still running. When it finishes, the per-batch partials are combined and copied into `results/allthebacteria/` with `bash run.sh combine $R/partials_stageD _stageD; bash run.sh fetch _stageD` in the P11 folder, which writes `atb_java_summary_stageD.json` and the `*_java_stageD.tsv` tables. The first AllTheBacteria tables (`insight*.tsv` without `_java`) were produced by a Python heuristic in Stage C and are not used here except for their alignment columns.

Table S16. Alignment survey of 72,533 AllTheBacteria genomes by phylum. Each ORF inherits the highest DIAMOND identity of its linclust representative (90% identity, 80% coverage) to the experimentally annotated Swiss-Prot training set of the CAFA (Kaggle) release (`blastp -sensitive -k 25`, E-value $\leq 10^{-3}$). Dark ORFs have less than 30% identity, including ORFs without any alignment (zero-alignment, percentage of dark ORFs in parentheses); twilight ORFs have 30–50% identity. Phyla with at least 15 genomes are shown individually. Only the alignment-derived columns of the source files are used. Sources: `results/allthebacteria/insight3b_phylum_summary.tsv` and `insight5b_phylum_nogd_vs_divergent.tsv` (alignment columns).

Domain	Phylum	Genomes	Species	ORFs	Dark ORFs (% of ORFs)	Zero-alignment (% of dark)	Twilight ORFs (% of ORFs)
Bacteria	Actinomycetota	31,304	2,212	135,787,329	49,575,316 (36.5%)	43,487,467 (87.7%)	29,717,896 (21.9%)
Bacteria	Pseudomonadota	20,734	4,952	86,910,368	35,130,850 (40.4%)	29,885,814 (85.1%)	25,746,504 (29.6%)
Bacteria	Bacillota	12,675	2,591	38,427,622	18,751,008 (48.8%)	15,418,954 (82.2%)	14,699,113 (38.3%)
Bacteria	Bacteroidota	3,341	1,101	13,846,301	8,929,800 (64.5%)	7,603,725 (85.2%)	4,147,192 (30.0%)
Bacteria	Unclassified_Bacteria	3,256	1	13,334,981	5,073,736 (38.0%)	4,395,885 (86.6%)	3,201,322 (24.0%)
Bacteria	Campylobacterota	327	176	613,728	316,688 (51.6%)	263,041 (83.1%)	234,298 (38.2%)
Bacteria	Spirochaetota	216	116	686,593	417,557 (60.8%)	356,510 (85.4%)	227,133 (33.1%)
Bacteria	Cyanobacteriota	115	88	636,008	387,725 (61.0%)	345,192 (89.0%)	192,362 (30.2%)
Bacteria	Desulfobacterota_I	72	51	238,413	122,590 (51.4%)	103,429 (84.4%)	93,454 (39.2%)
Bacteria	Fusobacteriota	67	33	156,125	84,893 (54.4%)	70,011 (82.5%)	57,616 (36.9%)
Bacteria	Deinococcota	65	44	240,057	127,706 (53.2%)	112,369 (88.0%)	92,949 (38.7%)
Bacteria	Desulfobacterota	51	47	183,894	94,154 (51.2%)	78,346 (83.2%)	71,363 (38.8%)
Bacteria	Verrucomicrobiota	48	36	178,665	108,910 (61.0%)	94,567 (86.8%)	58,263 (32.6%)
Bacteria	Chlamydiota	39	15	52,103	28,715 (55.1%)	24,523 (85.4%)	18,784 (36.1%)
Bacteria	Planctomycetota	31	22	179,627	119,699 (66.6%)	107,190 (89.5%)	51,524 (28.7%)
Bacteria	Fibrobacterota	28	17	83,173	52,000 (62.5%)	45,178 (86.9%)	25,640 (30.8%)
Bacteria	Acidobacteriota	27	25	129,097	75,601 (58.6%)	65,274 (86.3%)	44,155 (34.2%)
Bacteria	Thermotogota	23	17	48,238	24,366 (50.5%)	19,076 (78.3%)	20,247 (42.0%)
Bacteria	Myxococcota	18	13	129,765	77,482 (59.7%)	69,470 (89.7%)	44,293 (34.1%)
Bacteria	Bacteroidota_A	15	11	44,235	23,386 (52.9%)	19,817 (84.7%)	16,872 (38.1%)
Mixed	17 further phyla (fewer than 15 genomes each)	81	64	226,824	125,395 (55.3%)	107,488 (85.7%)	82,323 (36.3%)
	All 37 phyla	72,533	11,632	292,133,146	119,647,577 (41.0%)	102,673,326 (85.8%)	78,843,303 (27.0%)

S8 Cross-clade perturbation under two injection designs

Supplementary Table S18 reports the nine-genome cross-clade perturbation experiment of the main text under both injection designs. Design v1 injects one claim per protein–term pair with a raw score drawn from $U(0.55, 0.85)$; design v2 draws the raw score from $U(0.62, 0.90)$, alternates the primary evidence source between sequence alignment and a neural channel, and adds a second corroborating claim (score $\max(0.52, \text{raw} - 0.12)$) to 60% of the pairs. Both designs inject 100 pairs per genome (50 anatomical and 50 taxon-restricted terms) with the same seeds. Rows with the integrator build of 2026-10-14 (`gspa-unified-bench.jar`, $\alpha_{\text{cons}} = 3.0$, $\alpha_{\text{anat}} = 3.5$) reproduce the first version of the experiment; rows with `gspa-unified.jar` use the v2.1.0 defaults ($\alpha_{\text{cons}} = \alpha_{\text{anat}} = 0.68$). Under design v1 the Stage 1 soft priors alone remove all but 2 of 900 injected claims, so Stage 2A has no measurable effect; design v2 is the one reported in the main text.

S9 Leave-subunit-out: exact McNemar tests

Supplementary Table S19 reports exact two-sided McNemar tests on strict Hit@1 (ties count as misses) for the leave-subunit-out benchmark under both decoy designs. The coherence-only prior gives the same per-trial Hit@1 outcome as the attenuated input, so the two comparisons against them coincide.

S10 Stage 3 benchmark v2: McNemar tests and trigger rates

Supplementary Table S20 reports exact two-sided McNemar tests of the guarded Stage 3 agent (seed 42) against every other configuration on the 80-anomaly benchmark and on the 50 ambiguous leave-reaction-out gaps. Supplementary Table S21 reports the Stage 3 trigger rates across the 88 multi-kingdom proteomes under the production cap of 150 anomalies per genome and without the cap.

Table S17. GSPA integrator over AllTheBacteria genomes. The Stage D snapshot covers every genome of the 15 wave-1 batches finished on 2026-10-19 (species representatives); run (a) covers the genomes that Stage C had already integrated, the first 20 by sample identifier in each of 230 batches. Both runs use `gspa-unified.jar` v2.1.0 with all six priors, three iterations, and Stage 2A relaxation, compared with Stage 0 noisy-OR on the same claims. A protein is rescued when it gains its first GO term with posterior ≥ 0.50; a pair is suppressed when its posterior falls below 0.50 or Stage 2A removes it. Rescues by prior count a protein under every prior with a positive entry for a rescued pair. Co-annotation conflicts include pairs that Stage 2A removed without a forbidden-table hit. Taxon-forbidden suppression requires the genome's taxid in the integrator's bundled NCBI hierarchy (1,302 taxa). The gap claims are KEGG-coverage gaps written by the batch pipeline. Runtime is the per-genome wall time of single-threaded JVM runs in Stage C; Stage D runs 96 genomes per multi-threaded JVM. Sources: `results/allthebacteria/atb_java_summary_stageD_snapshot.json`, `atb_java_summary.json`, `insight3_per_genome_java.tsv` (pipeline P11, `benchmark/allthebacteria/aggregate_atb_java.py`).

Quantity	**Stage D snapshot**	**Run (a)**
Batches	15	230
Genomes	7,009	4,592
Phyla / species	29/3,965	16/767
ORFs	26,080,797	17,986,002
Dark ORFs (< 30% identity)	12,908,595 (49.5%)	7,121,089 (39.6%)
of which zero-alignment	10,866,712 (41.7%)	6,085,468 (33.8%)
Twilight ORFs (30–50%)	8,971,460 (34.4%)	4,503,540 (25.0%)
Rescued proteins (first term ≥ 0.50)	15,603 (0.06%)	5,694 (0.03%)
dark	2,485 (0.02%)	833 (0.01%)
zero-alignment	0	0
twilight	12,520 (0.14%)	3,250 (0.07%)
by prior: essentiality	10,815	4,231
by prior: coherence	3,253	1,256
by prior: genomic context	1,894	248
by prior: Stage 2A relaxation	66	356
by prior: gap filling	0	0
Rescued protein–term pairs	86,073	54,034
Suppressed protein–term pairs	875,786	568,068
taxon-forbidden	284,481	184,327
genome-wide co-annotation conflict	521,301	352,533
other	70,004	31,208
Genomes with taxid in the bundled taxonomy	2,309 (32.9%)	1,966 (42.8%)
KEGG-coverage gap claims (gap-filling prior fired)	30,984 (0)	19,608 (0)
Median JVM time per genome (Stage 0 + full run)	–	235 s ($n = 4{,}570$)

Table S18. Cross-clade perturbation experiment under two injection designs and two integrator builds. In each of nine bacterial and archaeal reference proteomes 100 eukaryotic developmental and clade-restricted protein–term pairs drawn from the taxon and anatomy constraint tables are injected (900 pairs in total). Design v1 gives each pair one claim with raw score $U(0.55, 0.85)$; design v2 draws raw scores from $U(0.62, 0.90)$ and adds a second, corroborating claim from the other channel to 60% of the pairs (score $\max(0.52, \text{raw} - 0.12)$). Cells give the injected pairs surviving the 0.50 threshold, summed over the nine genomes (out of 900), with the mean experimental-annotation micro $F_{\max}$ over the nine perturbed proteomes in parentheses, for Stage 0 pooling (A1), the taxonomic and anatomical priors (A6), all Stage 1 priors (A7), and Stage 2A deletion (A8) or relaxation (A9). The first row reproduces the earlier run with its integrator build (`gspa-unified-bench.jar`) and raw claim protein identifiers; the other two rows use the current build (`gspa-unified.jar`) with normalised protein identifiers in the claims, which supersede the raw-identifier runs of the current build.

Injection Design	**Integrator (Claim IDs)**	**A1: Stage 0**	**A6: + B_3, B_4**	**A7: All Stage 1**	**A8: + Stage 2A Deletion**	**A9: + Stage 2A Relaxation**
v1: raw score $U(0.55, 0.85)$, single claim	`gspa-unified-bench.jar` (raw IDs)	128 (0.2852)	2 (0.2894)	2 (0.2846)	2 (0.2846)	2 (0.2810)
v1: raw score $U(0.55, 0.85)$, single claim	`gspa-unified.jar` (normalised IDs)	149 (0.2824)	2 (0.2851)	10 (0.2828)	2 (0.2852)	2 (0.2863)
v2: raw score $U(0.62, 0.90)$, 60% corroborated	`gspa-unified.jar` (normalised IDs)	735 (0.2819)	137 (0.2849)	137 (0.2826)	24 (0.2852)	24 (0.2863)

Table S19. Exact McNemar tests for the leave-subunit-out benchmark. Two-sided exact binomial tests on the discordant trials of the strict Hit@1 indicator (true subunit uniquely ranked first), per decoy design and cohort; b counts trials that only configuration A ranks correctly and c trials that only configuration B ranks correctly.

Decoy Design	Cohort	Comparison (A vs B)	Trials (n)	b (A only)	c (B only)	p
unrelated loci	All 6 species	coherence + operon vs attenuated input	175	69	3	2.6×10^{-17}
unrelated loci	All 6 species	coherence + operon vs coherence prior only	175	69	3	2.6×10^{-17}
unrelated loci	All 6 species	full GSPA (Stage 3) vs coherence + operon	175	3	1	0.625
unrelated loci	Out-of-sample (5 species)	coherence + operon vs attenuated input	145	56	2	1.2×10^{-14}
unrelated loci	Out-of-sample (5 species)	coherence + operon vs coherence prior only	145	56	2	1.2×10^{-14}
unrelated loci	Out-of-sample (5 species)	full GSPA (Stage 3) vs coherence + operon	145	3	1	0.625
unrelated loci	*Synechocystis* PCC 6803	coherence + operon vs attenuated input	25	4	1	0.375
unrelated loci	*Synechocystis* PCC 6803	coherence + operon vs coherence prior only	25	4	1	0.375
unrelated loci	*Synechocystis* PCC 6803	full GSPA (Stage 3) vs coherence + operon	25	2	0	0.500
within operon	All 6 species	coherence + operon vs attenuated input	175	7	12	0.359
within operon	All 6 species	coherence + operon vs coherence prior only	175	7	12	0.359
within operon	All 6 species	full GSPA (Stage 3) vs coherence + operon	175	32	12	3.7×10^{-3}
within operon	Out-of-sample (5 species)	coherence + operon vs attenuated input	145	7	12	0.359
within operon	Out-of-sample (5 species)	coherence + operon vs coherence prior only	145	7	12	0.359
within operon	Out-of-sample (5 species)	full GSPA (Stage 3) vs coherence + operon	145	22	12	0.121
within operon	*Synechocystis* PCC 6803	coherence + operon vs attenuated input	25	1	2	1.000
within operon	*Synechocystis* PCC 6803	coherence + operon vs coherence prior only	25	1	2	1.000
within operon	*Synechocystis* PCC 6803	full GSPA (Stage 3) vs coherence + operon	25	3	3	1.000

Table S20. Exact McNemar tests for the Stage 3 benchmark. The guarded multi-turn agent (seed 42, temperature 0) is compared with every other configuration on the 80-anomaly benchmark (correct final action) and on the LRO-50 gaps (Hit@1). b counts items only the agent gets right and c items only the comparison method gets right; p is the two-sided exact binomial test on the discordant items.

Benchmark	Comparison Method (B)	Items (n)	Correct (Agent / B)	b	c	p
80 anomalies	Stage 1 + Stage 2 Deterministic Only	80	62/40	32	10	9.4×10^{-4}
80 anomalies	Stage 3 Agent – Fast 1-Step Mode	80	62/50	19	7	0.029
80 anomalies	Stage 3 Agent – Qwen 27B + Guardrail (seed=43)	80	62/57	8	3	0.227
80 anomalies	Stage 3 Agent – Qwen 27B + Guardrail (seed=44)	80	62/53	11	2	0.022
80 anomalies	Stage 3 Agent – Qwen 27B Without Guardrail	80	62/48	19	5	6.6×10^{-3}
80 anomalies	Direct Zero-Shot LLM (No Tools, No Guardrail)	80	62/22	47	7	2.3×10^{-8}
80 anomalies	Posterior Threshold ($\geq$0.35) + Guardrail	80	62/40	28	6	2.0×10^{-4}
80 anomalies	Logistic Regression (5-Fold Stratified CV, 3 Pre-Tool Scalars)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	Logistic Regression (5-Fold Stratified CV, 3 Pre-Tool Scalars) + Guardrail	80	62/60	12	10	0.832
80 anomalies	Random Forest (5-Fold Stratified CV, 3 Pre-Tool Scalars)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	Random Forest (5-Fold Stratified CV, 3 Pre-Tool Scalars) + Guardrail	80	62/60	12	10	0.832
80 anomalies	GBDT (5-Fold Stratified CV, 3 Pre-Tool Scalars)	80	62/38	33	9	2.7×10^{-4}
80 anomalies	GBDT (5-Fold Stratified CV, 3 Pre-Tool Scalars) + Guardrail	80	62/58	13	9	0.523
80 anomalies	MLP (5-Fold Stratified CV, 3 Pre-Tool Scalars)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	MLP (5-Fold Stratified CV, 3 Pre-Tool Scalars) + Guardrail	80	62/60	12	10	0.832
80 anomalies	Logistic Regression (5-Fold Stratified CV, 8 Post-Tool Features)	80	62/38	32	8	1.8×10^{-4}
80 anomalies	Logistic Regression (5-Fold Stratified CV, 8 Post-Tool Features) + Guardrail	80	62/58	12	8	0.503
80 anomalies	Random Forest (5-Fold Stratified CV, 8 Post-Tool Features)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	Random Forest (5-Fold Stratified CV, 8 Post-Tool Features) + Guardrail	80	62/60	12	10	0.832
80 anomalies	GBDT (5-Fold Stratified CV, 8 Post-Tool Features)	80	62/37	35	10	2.5×10^{-4}
80 anomalies	GBDT (5-Fold Stratified CV, 8 Post-Tool Features) + Guardrail	80	62/57	15	10	0.424
80 anomalies	MLP (5-Fold Stratified CV, 8 Post-Tool Features)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	MLP (5-Fold Stratified CV, 8 Post-Tool Features) + Guardrail	80	62/60	12	10	0.832
80 anomalies	Single Feature: current_posterior (argmax candidate, CV-tuned threshold)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	Single Feature: current_posterior (argmax candidate, CV-tuned threshold) + Guardrail	80	62/60	12	10	0.832
80 anomalies	Single Feature: final_logodds (argmax candidate, CV-tuned threshold)	80	62/40	32	10	9.4×10^{-4}
80 anomalies	Single Feature: final_logodds (argmax candidate, CV-tuned threshold) + Guardrail	80	62/60	12	10	0.832
80 anomalies	Single Feature: suggestion_score (argmax candidate, CV-tuned threshold)	80	62/38	33	9	2.7×10^{-4}
80 anomalies	Single Feature: suggestion_score (argmax candidate, CV-tuned threshold) + Guardrail	80	62/58	13	9	0.523
LRO-50	Single Feature: RLGC Suggestion Score (Rank Candidates by suggestion_score)	50	42/17	25	0	6.0×10^{-8}
LRO-50	Single Feature: Stage 1 Final Log-Odds (Rank Candidates by final_logodds)	50	42/36	11	5	0.210
LRO-50	Single Feature: ESM-2 Reaction-Centroid Cosine (Rank Candidates by esm2_centroid calibrated_prob)	50	42/48	1	7	0.070
LRO-50	RLGC ordered baseline (rlgc_ordered_pids)	50	42/17	25	0	6.0×10^{-8}
LRO-50	Logistic Regression (5-Fold CV, 3 Pre-Tool Scalars)	50	42/47	2	7	0.180
LRO-50	Random Forest (5-Fold CV, 3 Pre-Tool Scalars)	50	42/42	7	7	1.000
LRO-50	GBDT (5-Fold CV, 3 Pre-Tool Scalars)	50	42/45	4	7	0.549
LRO-50	MLP (5-Fold CV, 3 Pre-Tool Scalars)	50	42/48	1	7	0.070
LRO-50	Logistic Regression (5-Fold CV, 8 Post-Tool Features)	50	42/48	1	7	0.070
LRO-50	Random Forest (5-Fold CV, 8 Post-Tool Features)	50	42/46	3	7	0.344
LRO-50	GBDT (5-Fold CV, 8 Post-Tool Features)	50	42/46	3	7	0.344
LRO-50	MLP (5-Fold CV, 8 Post-Tool Features)	50	42/46	3	7	0.344
LRO-50	Direct Zero-Shot LLM (No Tools, No Guardrail) (seed=42)	50	42/20	23	1	3.0×10^{-6}
LRO-50	Stage 3 Agent – Fast 1-Step Mode (seed=42)	50	42/18	24	0	1.2×10^{-7}
LRO-50	Stage 3 Agent – Qwen 27B + Guardrail (seed=43)	50	42/43	3	4	1.000
LRO-50	Stage 3 Agent – Qwen 27B + Guardrail (seed=44)	50	42/44	1	3	0.625
LRO-50	Stage 3 Agent – Qwen 27B Without Guardrail (seed=42)	50	42/47	2	7	0.180

Table S21. Stage 3 trigger rates on 88 proteomes. Proteins that raise at least one Stage 3 anomaly under the production cap of 150 anomalies per proteome and without the cap (pooled percentage of all proteins), and anomalies per category (capped; uncapped in parentheses). The last column counts proteomes that reach the cap.

Domain	Proteomes	Proteins	Triggered (Cap 150)	Triggered (Uncapped)	Pathway Gap	Incomplete Complex	Borderline Essential	Borderline Conflict	Proteomes at Cap
Bacteria	47	94,666	4,902 (5.2%)	14,849 (15.7%)	3,826 (4,025)	2 (2)	45 (53)	3,119 (23,655)	45
Archaea	15	9,162	618 (6.7%)	845 (9.2%)	239 (239)	0 (0)	7 (7)	742 (1,281)	2
Eukaryota	6	5,546	362 (6.5%)	601 (10.8%)	30 (30)	4 (4)	5 (5)	734 (2,398)	5
Viruses	20	1,879	93 (4.9%)	93 (4.9%)	0 (0)	0 (0)	18 (18)	122 (122)	0
All	88	111,253	5,975 (5.4%)	16,388 (14.7%)	4,095 (4,294)	6 (6)	75 (83)	4,717 (27,456)	52